

MASTERING AZ-104

Microsoft Azure Administrator — Reordered Study Edition

MICROSOFT CERTIFICATION — BOOK 2 OF 4

A complete, hands-on course — from zero to administering Azure. Every exam objective explained in plain English, with analogies, portal and CLI walkthroughs, memory hooks, and practice questions.

Aligned to the official skills measured as of 17 April 2026.

EXAM TIP

REORDERED EDITION: chapters and sections in this copy have been resequenced to match the exact order of Microsoft's official AZ-104 skills-measured outline, so the table of contents below can be studied top-to-bottom as a direct checklist against the exam guide. See the Preface for a full list of what moved and why.

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IN PLAIN ENGLISH

Page numbers are approximate for this edition and will shift slightly depending on your PDF viewer's rendering — use the chapter/section numbers as your primary navigation, not the page count.

Preface: How to Use This Book

Reordered Edition — resequenced to match Microsoft's official AZ-104 skills-measured outline (17 April 2026), section for section, so you can study in the exact order the exam guide lists.

This edition keeps every word of the original book but changes the ORDER chapters and sections appear in, so the table of contents now walks line-by-line through Microsoft's official skill areas. Four changes were made:

What changed from the original edition

Change	Why
Storage chapters 4 and 5 swapped	The official outline lists "Configure access to storage" before "Configure and manage storage accounts"
Chapter 3 sections reordered (Policy → locks/tags → resource groups/subscriptions → costs → management groups)	Matches the official bullet order, with management groups moved to the end, exactly as listed
Chapter 8 sections reordered (create → encryption at host → moving VMs → availability → scale sets)	Matches the official bullet order for "Create and configure virtual machines"
New section 11.4 "Troubleshooting network connectivity" added	The official outline lists this as its own bullet under virtual networks, separate from load-balancer troubleshooting
Chapter 14 sections reordered (metrics/logs → alerts → Insights → Network Watcher)	Matches the official bullet order under "Monitor resources in Azure"

AZ-104 is a doing exam. This book teaches you the concepts and then shows you how each task is actually done — in the portal and on the command line — so the skills stick.

Unlike a fundamentals exam, AZ-104 expects you to implement and manage Azure, not just describe it. You'll meet interactive lab questions and case studies. So this book pairs every concept with how it works in practice: what you click in the Azure portal, and the equivalent Azure CLI or PowerShell command. You don't need prior Azure experience — Chapter 0 builds the foundation first.

The learning aids on every topic

IN PLAIN ENGLISH

Strips the jargon and says what a thing is in everyday words.

ANALOGY

Ties a technical idea to real life so it sticks.

```
# The exact portal path or command to perform a task
az group create --name myRG --location eastus
```

MEMORIZE THIS

The precise fact the exam tests — the numbers, limits, and distinctions.

EXAM TIP

The trick wordings and look-alikes that cost marks.

How to study for a hands-on exam

- Read for understanding first, then re-read the boxes and tables where the testable facts live.
- Do the labs. Get a free Azure account (\$200 credit for 30 days) and actually perform each task. AZ-104 rewards muscle memory.
- Learn portal AND CLI. Questions come both ways. When you read a command box, run it.
- Use the Quick Checks at the end of each chapter, and the practice bank in the appendix.
- Confirm the live objectives on Microsoft Learn before booking — this book follows the 17 April 2026 outline.

Orientation: The Azure Foundation

Chapter 0

What you'll learn

- How Azure is organized: tenants, subscriptions, resource groups, resources
- Regions, availability zones, and why location matters
- Azure Resource Manager, the portal, CLI, PowerShell, and Cloud Shell
- What the AZ-104 exam is and how it's scored

IN PLAIN ENGLISH

If you've never opened the Azure portal before, here's the one-sentence version of this whole book: Azure is a website full of "rent-a-computer-thing" services, and being an "Azure Administrator" means knowing where things live, how to create and connect them safely, and how to keep them running and affordable. Every chapter after this one is really just one category of "thing you can rent" (a server, a storage drive, a network) explained in depth. Nothing here assumes you've touched a cloud console before.

0.1 How Azure is organized: the management hierarchy

Before you can administer Azure, you need the map. Everything in Azure lives inside a hierarchy of containers. Understanding this hierarchy is the single most important foundation for the whole exam, because governance, billing, and access all flow through it.

Level	What it is
Management group	A container for multiple subscriptions, so you can apply governance to many at once
Subscription	A billing and access boundary; resources are billed to a subscription
Resource group (RG)	A folder that holds related resources sharing a lifecycle
Resource	An individual service instance — a VM, storage account, database, etc.

ANALOGY

Think of a company's filing system. The tenant is the whole company. Management groups are divisions. Subscriptions are departments with their own budgets. Resource groups are project folders. Resources are the actual documents. Rules set at a higher level flow down to everything beneath.

MEMORIZE THIS

The hierarchy, top to bottom: Management group → Subscription → Resource group → Resource. Governance and access assigned at a higher scope are inherited by everything below. A resource lives in exactly one resource group.

EXAM TIP

A resource group is a logical container, not a location boundary — resources in one RG can be in different regions. But the RG itself has a region (where its metadata lives).

0.2 Regions and availability zones

Azure runs in datacenters grouped into regions (like East US or West Europe) around the world. Within many regions are availability zones — physically separate datacenters with independent power and networking. Spreading resources across zones protects you if one datacenter fails. Some regions are paired with another region for disaster recovery.

MEMORIZE THIS

Region = a geographic set of datacenters. Availability zone = a physically separate datacenter within a region (protects against datacenter failure). Region pairs support cross-region recovery. Choose regions for latency, compliance/residency, and feature availability.

0.3 Azure Resource Manager and your admin tools

DEFINITION — Azure Resource Manager (ARM)

The deployment and management layer of Azure. Every request — from the portal, CLI, PowerShell, or a template — goes through ARM, which handles authentication, applies RBAC, and creates/updates resources.

You'll manage Azure through several interfaces, and the exam tests all of them:

Tool	What it is	Best for
Azure portal	The web GUI (portal.azure.com)	Learning, one-off tasks, visual work
Azure CLI	Cross-platform command line (az commands)	Scripting on Linux/macOS/Windows
Azure PowerShell	PowerShell module (Az cmdlets)	Windows-centric scripting/automation
Cloud Shell	Browser-based shell with CLI/PowerShell built in	Running commands without local setup
ARM templates / Bicep	Declarative infrastructure-as-code files	Repeatable, automated deployments

```
# Azure CLI: create a resource group
az group create --name myRG --location eastus
# PowerShell equivalent
New-AzResourceGroup -Name myRG -Location eastus
```

IN PLAIN ENGLISH

Whatever tool you use, it all funnels through Resource Manager. Learn the portal to understand concepts, then learn the CLI/PowerShell verbs because the exam will show you commands and ask what they do.

EXAM TIP

Know the command verbs: Azure CLI uses `az <service> create/list/show/delete`; PowerShell uses `Verb-AzNoun` (New-, Get-, Set-, Remove-). If you see `az`, it's CLI; if you see `-Az`, it's PowerShell.

0.4 About the AZ-104 exam

Fact	Detail
Full name	Microsoft Certified: Azure Administrator Associate
Level	Associate (Microsoft recommends ~6 months hands-on Azure)
Format	40–60 questions: multiple choice, drag-and-drop, case studies, and interactive labs
Time	100 minutes
Passing score	700 / 1000 (scaled)
Cost	~\$165 USD (varies by region)
Validity	1 year; renew free online via Microsoft Learn
Latest update	17 April 2026 (Entra naming, Container Apps, Bicep, deployment focus)

The five skill areas and how this book maps to them:

Part	Skill area	Weight	Chapters
1	Manage Azure identities and governance	20–25%	1–3
2	Implement and manage storage	15–20%	4–6
3	Deploy and manage Azure compute resources	20–25%	7–10
4	Implement and manage virtual networking	15–20%	11–13
5	Monitor and maintain Azure resources	10–15%	14–15

EXAM TIP

Identities/governance and compute are the two heaviest areas (up to 25% each) — together nearly half the exam. But networking and storage are dense and detail-heavy, and the interactive labs often come from compute and networking. Don't skip anything.

0.5 How AZ-104 questions actually get tricky

Having now seen real exam material, a pattern is worth naming up front: AZ-104 rarely asks "what does X do?" in isolation. It asks you to pick between two or three plausible-sounding options that all sound like they could work, where only one satisfies every constraint in the scenario. The details that decide the answer are almost always exact numbers (SLA percentages, priority ranges, retention days, IP counts) or a precise default behavior (what happens if you don't configure something). Every chapter from here on adds a "Sharpen the Edge" section with exactly these numbers and defaults, verified against current Microsoft documentation.

MEMORIZE THIS

When two answers both sound technically correct, look for the most specific constraint in the question — a region, a cost requirement, an exact number, or a phrase like "without downtime" or "without opening inbound ports." That constraint is almost always what eliminates the wrong-but-plausible option.

QUICK CHECK

Cover the answers. Say your answer out loud, then reveal.

Q1. List the Azure management hierarchy from top to bottom.

A. Management group → Subscription → Resource group → Resource.

Q2. What does Azure Resource Manager do?

A. It's the management layer every request passes through — handling auth, RBAC, and resource deployment.

Q3. What's the difference between a region and an availability zone?

A. A region is a geographic set of datacenters; an availability zone is a physically separate datacenter within a region.

Q4. You see the command 'New-AzVM'. Is that CLI or PowerShell?

A. PowerShell (Verb-AzNoun). Azure CLI would be 'az vm create'.

CHAPTER SUMMARY — KEY TAKEAWAYS

- Azure hierarchy: management group → subscription → resource group → resource; higher scopes are inherited.
- Regions group datacenters; availability zones are separate datacenters within a region; region pairs aid recovery.
- Everything flows through Azure Resource Manager; manage via portal, CLI (az), PowerShell (Az), Cloud Shell, or templates/Bicep.
- AZ-104 is a 100-minute associate exam with interactive labs; pass at 700/1000. Identities/governance and compute are heaviest.

Identities & Governance

Microsoft Entra Users & Groups

Chapter 1

What you'll learn

- Create and manage users and groups in Microsoft Entra ID
- The difference between member and guest (external) users
- Assigned vs dynamic group membership
- Managing licenses and configuring self-service password reset

IN PLAIN ENGLISH

Zoom out for a second: nothing in Azure can happen without someone (or something) signing in first. This whole chapter is really about answering two everyday-office questions in the cloud: "who works here?" (users) and "what team are they on?" (groups). Everything else — licenses, guest access, passwords — is just the practical mechanics of managing those two things at scale.

1.1 Users: members and guests

Every person who signs in to your Azure/Microsoft 365 environment needs an identity in Microsoft Entra ID. There are two broad kinds. A member is a normal account that belongs to your organization (your employees). A guest is an external person invited to collaborate — their account really lives in another directory, and you're just inviting them in.

User type	Belongs to	Created by
Member	Your organization's tenant	Created directly, or synced from on-prem AD
Guest (external)	Another directory / email	Invited via B2B collaboration (UPN shows #EXT#)

```
# Create a user with Azure CLI
az ad user create --display-name Jo --user-principal-name jo@contoso.com \
--password P@ssw0rd!
# Bulk-create users in the portal via CSV import
```

MEMORIZE THIS

Member = internal account. Guest = external collaborator invited via B2B; their UPN contains #EXT#. You can create users individually, in bulk via CSV, or by syncing on-prem AD.

Managing user and group properties

Creating an account is only the start — day to day administration is mostly about editing properties on the users and groups you already have. On a user object you'll routinely update profile fields (job title, department, manager, contact info), enable or disable sign-in (block sign-in without deleting the account), and reset or set a temporary password. On a group you'll update its name, description, owners (who can manage membership),

and add or remove members.

```
# Update a user's job title and department (CLI)
az ad user update --id jo@contoso.com \
--job-title "Support Analyst" --department "Helpdesk"
# Block a user from signing in without deleting them
az ad user update --id jo@contoso.com --account-enabled false
```

MEMORIZE THIS

Disabling sign-in (`account-enabled=false`) blocks access while preserving the account, mailbox, and group memberships — the standard first step for an employee on leave, versus deleting the user which is destructive and used for offboarding.

1.2 Groups: assigned vs dynamic

Managing access one person at a time doesn't scale, so you put users into groups and grant access to the group. Two things to know: the group type and the membership style.

Choice	Options	Meaning
Group type	Security	Grants access to resources and roles
	Microsoft 365	Adds shared mailbox/SharePoint for collaboration
Membership	Assigned	You add/remove members manually
	Dynamic	Members are added automatically by a rule on their attributes

ANALOGY

A dynamic group is like a smart playlist: you define a rule ("all songs tagged jazz") and the list fills itself. Change a song's tag and it joins or leaves automatically — just like a user moving departments.

```
# Example dynamic membership rule (users in Sales)
user.department -eq "Sales"
```

MEMORIZE THIS

Security groups grant access; Microsoft 365 groups add collaboration. Dynamic membership uses attribute rules and requires Entra ID P1. Dynamic user and dynamic device groups are separate — one group can't contain both.

1.3 Licenses and self-service password reset

Features like Entra ID P1/P2 and Microsoft 365 apps are unlocked by licenses assigned to users — directly or, more efficiently, through group-based licensing (assign the license to a group and every member inherits it). Self-service password reset (SSPR) lets users reset their own passwords after registering authentication

methods, cutting help-desk load.

MEMORIZE THIS

Group-based licensing assigns licenses to a group so members inherit them. SSPR requires users to register methods first; you choose how many methods are required and which are allowed. SSPR for all users requires Entra ID P1/P2.

1.4 Sharpen the edge: precise details that decide tricky questions

Detail	Exact behavior
License conflicts	If a user has two licenses granting the same service plan (e.g. Exchange Online from two SKUs), Azure disables the duplicate service plan automatically — it does not block the assignment or throw an error
Group-based licensing errors	If a group license can't apply to a member (e.g. a usage-location mismatch, a required field missing), that ONE user shows a licensing error while the rest of the group is unaffected
Removing a user from a licensed group	Immediately removes the inherited license and any data tied to it starts its normal grace/retention clock — the same as a direct unassignment
Bulk CSV operations	The admin center's bulk-create/bulk-invite templates are fixed CSV schemas — you cannot add custom columns; anything beyond the template's fields requires PowerShell/Graph
Dynamic group processing	A dynamic membership rule change is NOT instant — Azure evaluates and updates membership on its own schedule, so a test immediately after a rule change can show stale results

EXAM TIP

A classic trap scenario: "A user reports Exchange Online isn't working even though two licenses are assigned." The answer is almost never "assign a third license" — it's checking for a duplicate/conflicting service plan that got auto-disabled, or a missing usage location (a license can't assign at all without one set on the user).

QUICK CHECK

Cover the answers. Say your answer out loud, then reveal.

Q1. An invited partner's account shows #EXT# in the UPN. What kind of user is it?

A. A guest (external) user, added via B2B collaboration.

Q2. You want a group whose membership updates automatically when someone joins Sales. What do you use?

A. A dynamic group with a membership rule like user.department -eq "Sales" (requires P1).

Q3. What must a user do before they can use SSPR?

A. Register the required authentication methods.

CHAPTER SUMMARY — KEY TAKEAWAYS

- Members are internal; guests are external B2B collaborators (#EXT# in UPN).
- Group type: Security (access) vs Microsoft 365 (collaboration). Membership: assigned vs dynamic (P1).
- Group-based licensing scales license assignment; SSPR lets users reset passwords after registering methods.

Azure RBAC: Access to Resources

Chapter 2

What you'll learn

- What Azure role-based access control is and how it works
- The core built-in roles and what each allows
- How role assignments work at different scopes and how inheritance flows
- How to interpret a user's effective access across scopes and groups

IN PLAIN ENGLISH

If Chapter 1 was about WHO exists (users and groups), this chapter is about WHAT they're allowed to touch once they're signed in. Having an account doesn't mean you can do anything with it — by default a brand-new user can see almost nothing. RBAC (role-based access control) is simply the system for saying "this person, this level of power, on this specific set of resources." It sounds abstract at first, but every single permission decision in Azure eventually comes down to this one mechanism.

2.1 How Azure RBAC works

DEFINITION — Azure RBAC

Role-based access control for Azure resources: you grant access by creating a role assignment, which binds a security principal (user, group, or service principal) to a role definition (a set of allowed actions) at a scope (the level it applies to).

Every RBAC grant is three things answered together: who (the principal), what they can do (the role), and where (the scope). Miss any one and the grant makes no sense.

ANALOGY

A role assignment is like a building pass. The person is the principal, the pass type (visitor, staff, manager) is the role, and the floors it opens are the scope.

2.2 The core built-in roles

Role	What it can do
Owner	Full access to everything, INCLUDING granting access to others
Contributor	Create and manage all resources, but CANNOT grant access to others
Reader	View resources only; no changes
User Access Administrator	Manage user access (role assignments) only

MEMORIZE THIS

Contributor can do almost anything EXCEPT assign roles. Only Owner and User Access Administrator can grant access to others. This single distinction is one of the most tested facts on the exam.

2.3 Scopes and inheritance

You can assign a role at four scope levels, matching the management hierarchy. Crucially, a role assigned at a higher scope is inherited by everything beneath it.

Scope (broadest first)	Assigning here grants access to...
Management group	All subscriptions and resources under it
Subscription	All resource groups and resources in it
Resource group	All resources in that group
Resource	Just that single resource

```
# Assign Reader at a resource group scope (CLI)
az role assignment create --assignee jo@contoso.com \
--role Reader --resource-group myRG
```

EXAM TIP

RBAC is additive — your effective access is the sum of all your assignments. But a deny assignment overrides allow. Assign at the narrowest scope that does the job (least privilege). Reader at a subscription means Reader on everything inside it.

2.4 Interpreting access assignments

Because RBAC is additive and inherited, a person's effective access to a resource is rarely a single assignment — it's the union of every assignment on that resource, its resource group, its subscription, and any management group above it, plus every group they belong to. Reading this correctly is its own exam skill.

The Azure portal's Access control (IAM) blade has a "View my access" and "Check access" tool that resolves this automatically for a given user and resource — use it instead of manually tracing every scope by hand.

```
# List all role assignments a user has, across scopes (CLI)
az role assignment list --assignee jo@contoso.com --all
# List assignments scoped to one resource group only
az role assignment list --resource-group myRG
```

Worked example

Jo is a member of the "Sales" group, which has Reader on the subscription. Jo personally also has Contributor assigned directly on one resource group, "RG-Sales." What is Jo's effective access? - On any resource inside RG-Sales: Reader (inherited from the group, via the subscription) + Contributor (direct) → the union is Contributor (the more permissive of the two applies). - On any other resource group in the subscription: only Reader (from the group assignment) applies.

MEMORIZE THIS

Effective access = the union of every assignment reachable through direct grants, group membership, and scope inheritance. A narrower, more permissive assignment doesn't get overridden by a broader, less permissive one — they simply add together (unless a deny assignment is present, which always wins).

2.5 Azure RBAC vs Microsoft Entra roles

Two different role systems exist, and they're easy to confuse. Azure RBAC controls access to Azure resources (VMs, storage). Entra roles control administration of the directory (users, groups, Entra settings).

MEMORIZE THIS

Azure RBAC = resources (Owner/Contributor/Reader at mgmt group/sub/RG/resource). Entra roles = directory objects (Global Administrator, User Administrator). Global Administrator is an Entra role, not an Azure RBAC role.

2.6 Sharpen the edge: more built-in roles and custom role structure

Beyond Owner/Contributor/Reader, Azure ships dozens of narrower, service-specific built-in roles — the exam expects you to recognize a few of the most common ones by name.

Built-in role	Scope of power
Virtual Machine Contributor	Manage VMs, but cannot manage the VNet or storage account they depend on
Storage Account Contributor	Manage storage accounts, but cannot access the data inside via RBAC alone (needs a data-plane role too)
Network Contributor	Manage networking resources (VNets, NSGs, public IPs), not VMs or storage
Backup Contributor	Manage Azure Backup (create vaults, policies, trigger backup/restore)
Security Reader	Read-only access to security posture in Defender for Cloud, no ability to change anything

EXAM TIP

These narrow roles are a common trap: a question gives someone "Virtual Machine Contributor" and then asks if they can also create a new VNet for the VM — they can't. Service-specific Contributor roles are scoped tightly to that one service.

Custom role structure

A custom role definition (in JSON) has four key sections: Actions (allowed management operations, using wildcards like Microsoft.Compute/virtualMachines/*), NotActions (exclusions carved out of Actions — NOT a deny, just a subtraction), DataActions/NotDataActions (the same idea for data-plane operations, like reading blob contents), and AssignableScopes (which subscriptions/RGs the role can even be assigned at).

MEMORIZE THIS

NotActions is not a Deny. If a user has a second role granting a NotActions-excluded permission through a different assignment, they still get it — NotActions only carves permissions out of THAT SPECIFIC role definition, it doesn't block them tenant-wide the way an explicit Deny assignment does.

QUICK CHECK

Cover the answers. Say your answer out loud, then reveal.

Q1. Which built-in role can manage all resources but not grant access to others?

A. Contributor.

Q2. You assign Reader at the subscription scope. What can the user see?

A. Every resource group and resource in that subscription (inheritance).

Q3. A user needs to manage VMs and also assign roles on them. Which single role fits?

A. Owner (Contributor can't assign roles).

Q4. Is 'Global Administrator' an Azure RBAC role or an Entra role?

A. An Entra (directory) role.

Q5. A user has Reader via a group at the subscription and Contributor directly on one resource group. What's their effective access in that RG?

A. Contributor — effective access is the union of all reachable assignments, and the more permissive one applies.

CHAPTER SUMMARY — KEY TAKEAWAYS

- A role assignment = principal (who) + role definition (what) + scope (where).
- Owner = full + can grant access; Contributor = manage all but can't grant access; Reader = view.
- Roles inherit down the hierarchy; RBAC is additive; deny overrides allow; use least privilege.
- Azure RBAC governs resources; Entra roles govern directory administration.

Subscriptions & Governance

Chapter 3 — reordered to follow the official bullet sequence: Azure Policy, then locks and tags, then resource groups/subscriptions, then cost management, then management groups.

What you'll learn

- Enforcing rules with Azure Policy
- Protecting resources with locks and organizing them with tags
- Organizing resources with resource groups and subscriptions
- Controlling spend with Cost Management, budgets, and Advisor
- Applying governance at scale with management groups

IN PLAIN ENGLISH

"Governance" is just a formal word for "how do we stop this from turning into a mess as more people and resources get added." Without it, anyone with access could create resources anywhere, in any region, with any name, and spend unlimited money with no one noticing until the bill arrives. This chapter is the set of guardrails — organizational, rule-based, and financial — that keep a growing Azure environment sane.

3.1 Azure Policy

DEFINITION — Azure Policy

A governance service that evaluates your resources against rules and either audits non-compliant ones or prevents non-compliant deployments outright.

Where RBAC controls who can act, Azure Policy controls what is allowed to exist. Typical policies: allow only certain regions, require a tag, restrict VM sizes, or enforce encryption. Policies have effects such as Audit (flag it), Deny (block it), or DeployIfNotExists (auto-remediate).

ANALOGY

RBAC is the bouncer checking IDs (who gets in). Azure Policy is the dress code (what's allowed once you're in) — even an Owner can be stopped from creating a resource that violates policy.

MEMORIZE THIS

RBAC = who can act; Azure Policy = what is allowed. Common effects: Audit (report), Deny (block deployment), DeployIfNotExists (remediate). Group policies into initiatives to manage them together.

3.2 Resource locks and tags

Governance tool	What it does
Lock: CanNotDelete	Resource can be modified but not deleted
Lock: ReadOnly	Resource can be read but not modified or deleted

Governance tool	What it does
Tags	Key/value labels (e.g. env=prod, costcenter=42) for organizing and cost reporting

EXAM TIP

Locks are inherited from parent scopes and apply on top of RBAC — even an Owner can't delete a CanNotDelete-locked resource until the lock is removed. Tags are NOT inherited by default (a resource doesn't automatically get its RG's tags).

3.3 Resource groups, subscriptions, and management groups

These containers (from Chapter 0) are also governance tools. You group resources that share a lifecycle into a resource group, and separate workloads or billing into subscriptions.

MEMORIZE THIS

A resource belongs to exactly one resource group; deleting the RG deletes all its resources. You can move resources between RGs and subscriptions.

3.4 Cost management

Cloud spend can surprise you, so Azure gives you tools to watch and cap it. Cost Management analyzes spend; budgets set thresholds that trigger alerts; and Azure Advisor recommends ways to cut cost (and improve security, reliability, and performance).

MEMORIZE THIS

Budgets trigger alerts at spend thresholds (they notify — they don't auto-stop resources). Azure Advisor gives recommendations across cost, security, reliability, performance, and operational excellence. Use tags to break down cost by project/department.

3.5 Management groups

Roll many subscriptions under management groups to apply Azure Policy and RBAC once, everywhere, instead of repeating the same configuration on every subscription individually.

MEMORIZE THIS

Management groups let you apply Azure Policy and RBAC across many subscriptions at once. This is the broadest scope in the hierarchy and the last piece needed to govern an entire tenant consistently.

3.6 Sharpen the edge: policy effects and lock/tag precision

"Audit, Deny, DeployIfNotExists" only scratches the surface of policy effects — the exam expects you to distinguish a few more.

Policy effect	What it does
Deny	Blocks the non-compliant resource from being created/updated at all
Audit	Allows the resource, but flags it as non-compliant in the compliance report
Append	Adds a specified field/value to the resource during creation (e.g. force a tag) without blocking it
Modify	Adds, updates, or removes properties/tags on existing resources during creation or update
DeployIfNotExists (DINE)	After a resource is created, automatically deploys a related resource if it's missing (e.g. auto-enable a Defender for Cloud plan)
Disabled	The policy exists but does nothing — used for staged rollouts or testing

EXAM TIP

Append vs Modify is a common trap: Append only works at creation time and can't touch existing resources; Modify can update properties on resources that already exist. If the requirement is "add this tag to all resources going forward," Append is enough; if it's "add this tag to everything, including what's already deployed," you need Modify (often paired with a remediation task).

Locks and tags, precisely

- A lock applied at a resource group is inherited by every resource inside it — even resources added later.
- CanNotDelete still allows the resource to be modified/scaled; ReadOnly blocks modification too, including scaling a VM or resizing a disk.
- Tags are NOT inherited by default — tagging a resource group does not tag the resources inside it (a Policy with the Modify/Append effect is how you'd force that).
- A single resource can have up to 50 tags; each tag name is limited to 512 characters and each value to 256 characters.

QUICK CHECK

Cover the answers. Say your answer out loud, then reveal.

Q1. You must ensure no VM larger than a certain size is ever created. RBAC or Azure Policy?

A. Azure Policy (with a Deny effect) — it controls what's allowed to exist.

Q2. Which lock lets you change a resource but not delete it?

A. CanNotDelete.

Q3. Does a budget automatically shut down resources when exceeded?

A. No — it triggers alerts/notifications; it doesn't stop resources.

Q4. How do you apply governance to many subscriptions at once?

A. Use a management group and assign policy/RBAC there.

CHAPTER SUMMARY — KEY TAKEAWAYS

- Azure Policy controls what's allowed (Audit/Deny/DeployIfNotExists); RBAC controls who can act.
- Locks (CanNotDelete/ReadOnly) protect resources and are inherited; tags organize and enable cost reporting.
- Resource groups, subscriptions, and management groups organize resources and carry governance.
- Cost Management + budgets + alerts track spend; Advisor recommends improvements; management groups apply policy/RBAC tenant-wide.

Storage

Securing Access to Storage

Chapter 4 — moved ahead of Storage Accounts & Redundancy to match the official outline's "Configure access to storage" bullet, which is listed first.

What you'll learn

- Access keys and why to protect them
- Shared access signatures (SAS) and stored access policies
- Identity-based access with Microsoft Entra
- Storage firewalls and virtual network restrictions

IN PLAIN ENGLISH

Your storage account is reachable from the internet by default, which means the only thing standing between your data and a stranger is how you control access to it. This chapter is a menu of four increasingly refined ways to answer "who's allowed to read or write this data?" — from a single all-powerful master key at one extreme, to precise, identity-based, time-limited permissions at the other. As a rule of thumb: the more precise the method, the more secure it is.

4.1 Access keys

Every storage account has two access keys that grant full control over all its data. They exist in pairs so you can rotate one while the other stays in use. Because they're all-powerful, treat them like the crown jewels — and prefer more limited methods where possible.

MEMORIZE THIS

Access keys give full access to the whole account. There are two so you can rotate without downtime. Never embed them in code; store them in Key Vault if you must use them.

4.2 Shared access signatures (SAS)

DEFINITION — Shared access signature (SAS)

A signed URL that grants limited, time-bound access to specific storage resources — you control which services, which permissions, and an expiry time.

ANALOGY

An access key is the master key to the whole building. A SAS token is a temporary visitor pass that opens only room 3, only until 5 PM, and only lets you look, not touch.

SAS concept	What to know
Scope	Limit to a service, container, or single object
Permissions	Read, write, list, delete — pick only what's needed

SAS concept	What to know
Expiry	Set a start/end time so access auto-revokes
Stored access policy	A server-side policy a SAS references, so you can revoke/change many SAS at once

EXAM TIP

SAS vs access key: keys = full account access; SAS = scoped and time-limited. A raw SAS can't be revoked before expiry — unless it's tied to a stored access policy, which lets you revoke it centrally.

4.3 Identity-based access and network restrictions

The most secure approach avoids shared secrets entirely: grant access with Microsoft Entra ID + RBAC, so a user or managed identity is authorized by their identity, not a key. This applies to Blob Storage directly, and to Azure Files as well.

Identity-based access for Azure Files specifically

By default, an Azure Files SMB share is accessed with the storage account key — like one shared master password for the whole share. Identity-based access replaces that with real user identities, so you can set NTFS-style permissions per user and get proper audit trails, exactly like an on-prem file server.

Identity source	How it works
On-premises AD DS	Register the storage account with your on-prem AD; users authenticate with their existing domain identity over a hybrid connection
Microsoft Entra Kerberos	Cloud-only Kerberos authentication straight from Entra ID — no on-prem AD needed
Microsoft Entra Domain Services	A managed, cloud-hosted domain that provides Kerberos auth for Files

MEMORIZE THIS

Identity-based access for Azure Files lets users authenticate with their own AD/Entra identity (Kerberos) instead of the shared storage account key, enabling per-user NTFS-style permissions. Configure it once at the storage account level, then set share/directory/file permissions as you would on-prem.

Separately, storage firewalls restrict which networks can reach the account at all — you can allow only specific virtual networks or public IP ranges, and deny everything else by default once configured.

MEMORIZE THIS

Prefer Entra ID + RBAC (identity-based, no secrets) over keys/SAS — this is the usual "most secure" answer. Storage firewalls restrict access to chosen VNets/IP ranges (deny public access by default when configured).

4.4 Sharpen the edge: the three SAS types

"SAS" isn't one thing — there are three distinct kinds, and the exam expects you to pick the right one for a scenario.

SAS type	Signed with	Scope
User delegation SAS	Entra ID credentials (most secure — no account key involved at all)	Blob storage only; tied to an Entra identity's permissions
Service SAS	The storage account key	One specific service (Blob, Files, Queue, or Table)
Account SAS	The storage account key	Can span MULTIPLE services at once and can grant account-level operations (like changing CORS rules) that a service SAS can't

EXAM TIP

If a question stresses "no storage account key should ever be exposed, even indirectly," the answer is a user delegation SAS — it's the only type not signed with the account key. If the requirement spans multiple services (blob AND queue) in one token, that's an account SAS.

SAS permission letters

SAS permissions are often shown as a compact string of letters: read, write, delete, list, add, create, update, process. Grant only the letters actually needed — a read-only share should be issued with just r, nothing else.

QUICK CHECK

Cover the answers. Say your answer out loud, then reveal.

Q1. You need to give a partner read-only access to one container for 24 hours. What do you issue?

A. A shared access signature (SAS) scoped to that container, read-only, expiring in 24 hours.

Q2. How can you revoke many issued SAS tokens at once?

A. Tie them to a stored access policy and revoke/modify that policy.

Q3. What's the most secure way to let an app access blobs without storing secrets?

A. Use a managed identity with Entra ID + RBAC (identity-based access).

CHAPTER SUMMARY — KEY TAKEAWAYS

- Access keys = full account access (two keys for rotation); guard them, use Key Vault.
- SAS = scoped, time-limited signed URLs; stored access policies allow central revocation.
- Entra ID + RBAC is the most secure (no secrets); storage firewalls restrict access to chosen networks.

Storage Accounts & Redundancy

Chapter 5

What you'll learn

- What a storage account is and the services inside it
- Redundancy options: LRS, ZRS, GRS, GZRS, and read-access variants
- Storage encryption and object replication
- Moving data with Storage Explorer and AzCopy

IN PLAIN ENGLISH

Think of a storage account as an unlimited external hard drive that Microsoft looks after for you — it never runs out of physical space, it's reachable from anywhere with an internet connection, and Microsoft keeps multiple hidden copies of everything you put on it. The only real decisions you make are: how many copies, and how far apart should they be kept? That's genuinely most of what this chapter covers — the rest is naming and moving files around.

5.1 What is a storage account?

DEFINITION — Storage account

A container that provides access to Azure Storage data services — Blob (objects/files), Files (SMB file shares), Queue (messaging), and Table (NoSQL) — under one namespace, with shared settings like redundancy and encryption.

A storage account is the top-level unit you create; inside it live blobs, file shares, queues, and tables. Key settings chosen at creation include the performance tier (Standard vs Premium), the account kind, and — most importantly for the exam — the redundancy option.

5.2 Redundancy: how many copies, and where

Azure always keeps multiple copies of your data. The redundancy option decides how many and how far apart — which determines what kind of failure you can survive. Learn this ladder cold.

Option	Copies / spread	Survives
LRS	3 copies in one datacenter	Disk/hardware failure only
ZRS	3 copies across availability zones in the region	A whole datacenter/zone failure
GRS	LRS locally + 3 async copies in a paired region	A full regional outage (via failover)
GZRS	ZRS locally + async copies in a paired region	Zone failure AND regional outage

Option	Copies / spread	Survives
RA-GRS / RA-GZRS	GRS/GZRS + read access to the secondary region	Same, plus you can read the secondary anytime

ANALOGY

Think of backups of an important document. LRS = three copies in one drawer. ZRS = a copy in three different rooms. GRS = copies in another city too. RA- = you're allowed to read the out-of-city copy anytime, not just after a disaster.

MEMORIZE THIS

LRS=Local (1 datacenter), ZRS=Zones (1 region), GRS=Geo (2 regions), GZRS=zones + geo. The RA-prefix adds Read Access to the secondary. More resilience = more cost. Only geo options survive a regional outage.

5.3 Encryption and object replication

All Azure Storage is encrypted at rest by default using Microsoft-managed keys — you don't turn it on. You can optionally supply your own keys (customer-managed keys in Key Vault). Object replication asynchronously copies blobs from one storage account to another (e.g. to bring data closer to users in another region).

MEMORIZE THIS

Storage encryption at rest is always on (Microsoft-managed keys by default; customer-managed keys optional). Object replication copies block blobs between accounts asynchronously.

5.4 Moving data: Storage Explorer and AzCopy

Tool	What it's for
Azure Storage Explorer	A GUI app to browse and manage storage data visually
AzCopy	A command-line tool for fast, scriptable bulk copy/move of data

```
# Copy a local folder to a blob container with AzCopy
azcopy copy './data' 'https://acct.blob.core.windows.net/mycontainer' --recursive
```

5.5 Sharpen the edge: naming, tiers, and SLA precision

Detail	Exact rule
Storage account name	3–24 characters, lowercase letters and numbers ONLY (no hyphens/uppercase), globally unique across all of Azure

Detail	Exact rule
Performance tiers	Standard (HDD-backed, general purpose) vs Premium (SSD-backed, low-latency — required for disks needing consistent high IOPS)
Changing redundancy after creation	You can usually change LRS to ZRS after the fact, but converting TO GZRS/RA-GZRS sometimes requires support or a migration, depending on the account's history — always check before assuming it's a simple toggle
GRS failover	A GRS/GZRS failover to the secondary region is a MANUAL customer-initiated action (or Microsoft-initiated in a true disaster) — it does not happen automatically the moment the primary region has a blip
RA-GRS read endpoint	The secondary region's data is exposed on a separate read-only endpoint with a -secondary suffix — your app must be coded to use it; it isn't automatically used as a fallback

EXAM TIP

A frequent trap: "An app needs to keep working, unmodified, if the primary region goes down." RA-GRS alone does NOT satisfy this — the app must be written to detect the failure and switch to the -secondary endpoint, or you must perform an account failover. Redundancy protects the data; it doesn't automatically redirect your application's traffic.

QUICK CHECK

Cover the answers. Say your answer out loud, then reveal.

Q1. Which redundancy option protects against a full region going down but is the cheapest that does so?

A. GRS (geo-redundant).

Q2. Do you need to enable encryption at rest for a storage account?

A. No — it's on by default with Microsoft-managed keys.

Q3. Which tool would you script to bulk-upload thousands of files?

A. AzCopy.

CHAPTER SUMMARY — KEY TAKEAWAYS

- A storage account groups Blob, Files, Queue, and Table services under shared settings.
- Redundancy ladder: LRS → ZRS → GRS → GZRS; RA- adds read access to the secondary; only geo options survive regional loss.
- Encryption at rest is always on; object replication copies blobs between accounts.
- Storage Explorer = GUI; AzCopy = scriptable bulk transfer.

Blob Storage & Azure Files

Chapter 6

What you'll learn

- Creating blob containers and file shares
- Access tiers: Hot, Cool, Cold, and Archive
- Protecting data with soft delete, snapshots, and versioning
- Automating tiering with blob lifecycle management

IN PLAIN ENGLISH

This chapter is where you actually put files into your storage account. Two questions drive almost everything here: "what kind of thing am I storing" (a random collection of files, like Blob — or a proper shared folder people map as a drive, like Files) and "how often will I actually look at it" (accessed constantly, or dumped once and rarely touched again). Get those two answers right and the rest of the chapter's decisions follow naturally.

6.1 Blob containers and file shares

Service	What it is	Access via
Azure Blob Storage	Object storage for unstructured data (files, images, backups), held in containers	HTTPS/REST, SDKs, SAS
Azure Files	Fully managed SMB/NFS file shares you can mount like a network drive	SMB mount, mapped drive

IN PLAIN ENGLISH

Use Blob when apps read/write objects over HTTP (like a photo store or backup target). Use Files when you want a shared drive that Windows/Linux machines can mount the old-fashioned way.

6.2 Access tiers

Not all data is accessed equally often. Blob access tiers let you trade storage cost against access cost based on how frequently you touch the data.

Tier	Use for	Trade-off
Hot	Frequently accessed data	Highest storage cost, lowest access cost
Cool	Infrequent (stored $\geq$ 30 days)	Lower storage, higher access cost
Cold	Rare (stored $\geq$ 90 days)	Lower still

Tier	Use for	Trade-off
Archive	Almost never (stored ≥ 180 days)	Cheapest storage, but OFFLINE — must rehydrate to read

MEMORIZE THIS

Tiers hot to cold: Hot → Cool → Cold → Archive. Archive is offline — reading requires rehydration (hours) to Hot or Cool first. Colder tiers have minimum-stay periods and early-deletion penalties.

6.3 Protecting data: soft delete, snapshots, versioning

Feature	What it does
Soft delete (blobs/containers)	Keeps deleted data recoverable for a retention window
Blob versioning	Automatically keeps previous versions when a blob is overwritten
Snapshots	Read-only point-in-time copies of a blob
Azure Files soft delete/snapshots	Same protections for file shares

ANALOGY

Soft delete is your recycle bin; versioning is your document's "version history"; a snapshot is a photo of the file at a moment in time. Together they let you undo mistakes.

6.4 Blob lifecycle management

DEFINITION — Lifecycle management

Rules that automatically move blobs to cooler tiers or delete them based on age or last-access time — so you don't pay Hot prices for cold data.

```
# Example rule intent (set as JSON policy in the portal)
# If a blob hasn't been modified in 30 days -> move to Cool
# If not modified in 180 days -> move to Archive; after 365 -> delete
```

MEMORIZE THIS

Lifecycle management auto-tiers or deletes blobs by age/last-access, saving cost with no manual work. Pair it with the tiers: keep new data Hot, age it to Cool/Cold/Archive, then delete.

6.5 Sharpen the edge: minimum retention and rehydration priority

Tier	Minimum stay before moving out (early-deletion fee applies)
Cool	30 days
Cold	90 days
Archive	180 days

Moving a blob out of a tier before its minimum stay is up (e.g. moving a Cool blob to Hot after 10 days) doesn't fail — it succeeds, but Azure bills an early-deletion penalty as if the blob had stayed the full minimum period.

Rehydrating from Archive

Rehydration priority	Typical time
Standard	Up to 15 hours
High	Under 1 hour for blobs under 10 GB (best-effort, no guaranteed SLA)

EXAM TIP

A scenario that says "we need this archived file back within the hour for an urgent request" is pointing at High-priority rehydration — note it's best-effort, not a hard guarantee, which is itself sometimes the detail a question tests ("is rehydration time guaranteed?" — no).

QUICK CHECK

Cover the answers. Say your answer out loud, then reveal.

Q1. You need a shared drive several VMs can mount like a network folder. Blob or Files?

A. Azure Files.

Q2. Which tier is cheapest to store but must be rehydrated before reading?

A. Archive.

Q3. A blob was overwritten by mistake. Which feature lets you recover the previous content?

A. Blob versioning (or a snapshot / soft delete).

Q4. How do you automatically move 90-day-old data to a cheaper tier?

A. A blob lifecycle management rule.

CHAPTER SUMMARY — KEY TAKEAWAYS

- Blob = object storage in containers; Files = mountable SMB/NFS shares.
- Tiers Hot/Cool/Cold/Archive trade storage vs access cost; Archive is offline (rehydrate to read).
- Soft delete, versioning, and snapshots protect against accidental loss/overwrite.
- Lifecycle management auto-tiers and deletes blobs by age to save cost.

Compute

Infrastructure as Code: ARM & Bicep

Chapter 7

What you'll learn

- What infrastructure as code is and why it matters
- Reading and modifying ARM templates
- Reading and modifying Bicep files
- Deploying, exporting, and converting templates

IN PLAIN ENGLISH

Everything you've built so far in this book — users, storage accounts — you could do by clicking through the Azure portal. This chapter introduces a different way to build things: writing down what you want in a text file, then handing that file to Azure and letting it build exactly what's described. It feels like an extra step at first, but it's the difference between describing a room to a builder from memory every single time versus handing them a blueprint. This is one of the more abstract topics for a newcomer — take it slowly, the payoff is real.

7.1 Why infrastructure as code?

Clicking through the portal is fine for one resource, but not for building the same environment reliably ten times. Infrastructure as code (IaC) describes your resources in a text file you can version, review, and deploy repeatedly. Azure's native IaC formats are ARM templates and Bicep.

ANALOGY

The portal is cooking by taste each time; IaC is a written recipe. The recipe gives the same dish every time, can be shared, and improved without re-remembering the steps.

7.2 ARM templates vs Bicep

	ARM template	Bicep
Format	JSON — verbose, lots of braces	A clean, purpose-built language
Relationship	The underlying format Azure accepts	Transpiles (compiles) down to ARM JSON
Readability	Harder to read/write	Much easier
Behavior	Declarative & idempotent	Declarative & idempotent

DEFINITION — Idempotent

Deploying the same template repeatedly produces the same end state — existing correct resources are left as-is, not duplicated.

MEMORIZE THIS

Bicep transpiles to ARM JSON — same result, friendlier syntax. Both are declarative (you describe the desired end state) and idempotent (safe to re-run). You can decompile ARM to Bicep and build Bicep to ARM.

```
# A complete, tiny Bicep file that creates one storage account
param storageName string
param location string = resourceGroup().location
resource stg 'Microsoft.Storage/storageAccounts@2023-01-01' = {
  name: storageName
  location: location
  sku: { name: 'Standard_LRS' }
  kind: 'StorageV2'
}
```

IN PLAIN ENGLISH

That's genuinely the whole thing — nine short lines describe an entire storage account. The first two lines are parameters (blanks Azure will ask you to fill in when you deploy). The resource block is the actual thing being created, with its properties listed inside. Notice there's no step-by-step "first do this, then do that" instructions anywhere — you just describe the end result you want, and Azure figures out how to get there. That's what "declarative" means in practice.

Interpreting and modifying a template

Both formats describe the same four things: parameters (values supplied at deploy time), variables (computed values reused in the file), resources (what gets created), and outputs (values returned after deployment). Reading a template means finding these four blocks; modifying one usually means changing a resource's properties or adding a parameter.

```
# A minimal Bicep file defining a storage account
param storageName string
param location string = resourceGroup().location
resource sa 'Microsoft.Storage/storageAccounts@2023-01-01' = {
  name: storageName
  location: location
  sku: { name: 'Standard_LRS' }
  kind: 'StorageV2'
}
output storageId string = sa.id
```

Reading this: `storageName` is a required parameter (no default); `location` is optional and defaults to the resource group's region; the resource block declares one storage account with LRS redundancy; the output returns its resource ID after deployment. To modify it — say, to make it ZRS-redundant instead — you'd change only the `sku.name` line to `'Standard_ZRS'`; nothing else needs to change because the rest of the template is unaffected.

EXAM TIP

On the exam you may be shown a partial ARM/Bicep snippet and asked to fill in or correct one line — look for the resource type/apiVersion, the properties that map to what you'd configure in the portal, and whether a value should be a parameter (changes per deployment) or a fixed value.

7.3 Deploying, exporting, and converting

```
# Deploy a Bicep or ARM file to a resource group
az deployment group create --resource-group myRG --template-file main.bicep
# Convert an existing ARM template to Bicep
az bicep decompile --file template.json
# Export a deployed resource group as a template (portal: Export template)
```

EXAM TIP

Know the directions: decompile = ARM → Bicep; build = Bicep → ARM. Export template captures existing resources as a template you can reuse. Templates deploy at a scope (resource group, subscription, or management group).

7.4 Sharpen the edge: template structure and deployment modes

An ARM/Bicep template has four functional sections you should recognize on sight: parameters (values supplied at deploy time), variables (computed/reused values inside the template), resources (what actually gets created), and outputs (values returned after deployment, like a generated connection string).

Deployment mode	Behavior on resources NOT in the template but present in the resource group
Incremental (default)	Leaves them alone — only adds/updates what's in the template
Complete	DELETES them — the resource group ends up matching the template exactly

EXAM TIP

Complete mode is destructive and a favorite trap: a question describing unexpected resource deletions after a deployment is almost always pointing at someone using Complete mode by mistake. Incremental is the safe default and what az deployment group create uses unless you explicitly pass --mode Complete.

QUICK CHECK

Cover the answers. Say your answer out loud, then reveal.

Q1. What does Bicep compile down to?

A. ARM template JSON.

Q2. What does 'idempotent' mean for a deployment?

A. Re-running it yields the same end state without duplicating existing resources.

Q3. How do you turn an existing ARM template into Bicep?

A. Decompile it (az bicep decompile).

CHAPTER SUMMARY — KEY TAKEAWAYS

- IaC describes resources in versionable text for repeatable deployments.
- Bicep is friendlier and transpiles to ARM JSON; both are declarative and idempotent.
- Decompile ARM->Bicep, build Bicep->ARM, and Export template captures existing resources.

Virtual Machines & Availability

Chapter 8 — reordered to follow the official bullet sequence: create, then encryption at host, then moving VMs, then availability sets/zones, then scale sets.

What you'll learn

- Creating and sizing virtual machines and their disks
- Encryption at host
- Moving VMs between resource groups, subscriptions, and regions
- Keeping VMs available with availability sets and zones
- Scaling automatically with Virtual Machine Scale Sets

IN PLAIN ENGLISH

A virtual machine is the most familiar thing in this whole book: it's just a computer — with an operating system, a hard disk, a certain amount of CPU and memory — except it lives in a Microsoft datacenter instead of under your desk, and you can create, resize, or delete it in seconds instead of ordering hardware. This chapter is about creating that computer correctly, and then making sure a single hardware failure somewhere in the datacenter can't take your service down with it.

8.1 Creating VMs, sizes, and disks

A virtual machine is a rented computer in Azure. When you create one you choose its size (the family/SKU that sets CPU and memory), its disks, its network, and its region. Disks come as managed disks in tiers (Standard HDD, Standard SSD, Premium SSD, Ultra) trading cost against performance.

```
# Create a Linux VM with the CLI
az vm create --resource-group myRG --name myVM \
--image Ubuntu2204 --size Standard_B2s --admin-username azureuser \
--generate-ssh-keys
```

MEMORIZE THIS

VM size/SKU sets CPU + memory (resize when stopped). Managed disk tiers: Standard HDD → Standard SSD → Premium SSD → Ultra (increasing performance/cost). OS disk + optional data disks; temp disk is not persistent.

8.2 Encryption at host and moving VMs

Encryption at host encrypts VM disk data on the host itself — including the temp disk and caches, which regular Azure Disk Encryption doesn't cover. It's a checkbox at VM creation (or enabled afterward on a supported VM size).

MEMORIZE THIS

Encryption at host encrypts temp disk + caches on the host, in addition to OS/data disk encryption. It must be supported by the VM size/region, and is enabled per-subscription first.

Moving a VM

Moving a VM's resource group or subscription is a lightweight metadata operation — the VM and its associated resources (disks, NIC) are reassigned to the new container without being rebuilt. Moving a VM to a different region is not a simple move: the VM must be redeployed in the target region (commonly via Azure Resource Mover, or by recreating it from a snapshot/image/ARM export), because compute resources are region-bound.

Move type	How it's done	Downtime
To another resource group	Select resources -> Move -> choose target RG (same subscription)	None — metadata only
To another subscription	Same Move operation, target a different subscription	None — metadata only
To another region	Azure Resource Mover, or rebuild from image/snapshot/template	Requires redeployment

```
# Move resources to another resource group (CLI)
az resource move --destination-group newRG \
--ids /subscriptions/xxx/resourceGroups/oldRG/providers/.../myVM
```

MEMORIZE THIS

Move within/across subscriptions = fast, metadata-only, no downtime. Move across regions = not a true move — requires redeployment (e.g. Azure Resource Mover) since a VM is tied to the region it was created in. Not every resource type supports moving — check compatibility before planning one.

8.3 High availability: sets and zones

A single VM can go down during hardware failure or maintenance. Azure offers two ways to protect against that, and you must know the difference.

Option	Protects against	How
Availability Set	Hardware failures & maintenance WITHIN one datacenter	Spreads VMs across fault domains and update domains
Availability Zones	An entire datacenter failure	Spreads VMs across physically separate zones in the region

DEFINITION — Fault domain / Update domain

Fault domain: A group of hardware sharing power and network — a single point of physical failure. Update domain: a group rebooted together during planned maintenance.

MEMORIZE THIS

Availability Set = fault + update domains, protects within a datacenter (~99.95% SLA). Availability Zones = separate datacenters, protects across them (~99.99% SLA). You pick this at VM creation; you generally can't add a running VM to a set later.

8.4 Scaling with VM Scale Sets

A Virtual Machine Scale Set (VMSS) runs a group of identical VMs that automatically scale out (add instances) or in (remove them) based on rules or a schedule — ideal for elastic workloads behind a load balancer.

MEMORIZE THIS

VMSS = identical, auto-scaling VMs (scale by metric or schedule). Scale up = bigger VM size (vertical); scale out = more instances (horizontal, what VMSS does).

8.5 Sharpen the edge: the exact SLA numbers

These percentages are directly testable, and the trap is that the number depends on BOTH the disk type AND the availability configuration — not just one or the other.

Configuration	SLA
Single VM, Standard HDD	No SLA guarantee (Standard HDD alone doesn't qualify)
Single VM, Standard SSD	~99.5%
Single VM, Premium SSD (or Ultra Disk) for ALL disks	99.9%
2+ VMs in an Availability Set	99.95%
2+ VMs across 2+ Availability Zones	99.99%

MEMORIZE THIS

A single VM only reaches 99.9% if EVERY disk (OS + all data disks) is Premium SSD or Ultra Disk — mixing in even one Standard disk drops the single-VM SLA. Also remember: you can only place a VM into an availability set at creation time — you cannot add an already-running VM to one afterward.

EXAM TIP

Also testable: not every VM size supports Availability Zones in every region — a question might describe a zone-redundant design failing to deploy because the chosen VM size/region combination doesn't support zones, which is a real constraint to check, not just a configuration toggle.

QUICK CHECK

Cover the answers. Say your answer out loud, then reveal.

Q1. You need VMs to survive an entire datacenter outage in a region. Set or zones?

A. Availability Zones.

Q2. What does an availability set spread VMs across?

A. Fault domains and update domains, within a single datacenter.

Q3. Which scaling adds more identical VM instances automatically?

A. Scale out, using a Virtual Machine Scale Set.

Q4. How do you resize a VM to more CPU/RAM?

A. Change its size/SKU (scale up) — usually with the VM stopped.

CHAPTER SUMMARY — KEY TAKEAWAYS

- VM size sets CPU/RAM; managed disk tiers trade cost vs performance (HDD->SSD->Premium->Ultra).
- Encryption at host protects temp disk + caches; moving a VM within/across subscriptions is metadata-only, across regions requires redeployment.
- Availability Set = fault/update domains within a datacenter; Availability Zones = across datacenters.
- VMSS auto-scales identical VMs; scale up = bigger, scale out = more.

Containers in Azure

Chapter 9

What you'll learn

- What containers are and how they differ from VMs
- Storing images in Azure Container Registry
- Running containers with Azure Container Instances and Container Apps
- Sizing and scaling containers

IN PLAIN ENGLISH

If VMs from the last chapter felt familiar, containers might feel like a strange new idea — so here's the plainest possible version: a developer packages up their application AND everything it needs to run (specific software versions, settings, libraries) into one single file called an image. That image runs identically whether it's on the developer's laptop or in Azure, because it's not relying on whatever happens to already be installed on the machine — it brought its own. "Running a container" just means starting up that self-contained package.

9.1 Containers vs VMs

DEFINITION — Container

A lightweight, portable package holding an application plus everything it needs to run, sharing the host OS kernel rather than including a full operating system like a VM.

ANALOGY

A VM is a whole house (its own OS, furniture, plumbing). A container is an apartment in a shared building — it has what it needs but shares the building's foundation (the host OS), so it's smaller and starts faster.

9.2 The three container services

Service	What it does	Use when
Azure Container Registry (ACR)	Private store for your container images	You need somewhere to keep/build images
Azure Container Instances (ACI)	Runs a single container fast, serverless, no orchestration	Quick, simple, short-lived container tasks
Azure Container Apps	Serverless containers with autoscaling, microservices, revisions	Production apps needing scale without managing Kubernetes

```
# Push then run a container image
az acr build --registry myACR --image app:v1 .
az container create --resource-group myRG --name c1 \
--image myACR.azurecr.io/app:v1 --cpu 1 --memory 1.5
```

MEMORIZE THIS

ACR = image storage/registry. ACI = fastest way to run a single container (serverless, no orchestration). Container Apps = serverless containers with autoscaling and microservices (built on Kubernetes, but you don't manage it). Full Kubernetes control = AKS (not the light options).

EXAM TIP

For a simple one-off container -> ACI. For a scalable production app without running Kubernetes yourself -> Container Apps. Sizing is set by CPU/memory per container; Container Apps scales on rules (including scale-to-zero).

9.3 Sharpen the edge: ACR tiers and ACI restart policies

ACR tier	Key difference
Basic	Lower storage/throughput; fine for dev/test
Standard	More storage/throughput for production workloads
Premium	Adds geo-replication (one registry, multiple regions), content trust (image signing), and private endpoint support

MEMORIZE THIS

Only Premium ACR supports geo-replication. A question about serving the same container image with low latency from multiple regions, from one registry, is pointing at ACR Premium, not Standard.

ACI restart policies

Policy	Behavior
Always	Restarts the container no matter how it exits (default)
Never	Runs once and stops, regardless of exit code
OnFailure	Restarts only if the container exits with an error

MEMORIZE THIS

For a one-off batch job in ACI (run a script and stop), use Never or OnFailure — the default Always would keep re-running it forever.

QUICK CHECK

Cover the answers. Say your answer out loud, then reveal.

Q1. Where do you store your private container images in Azure?

A. Azure Container Registry (ACR).

Q2. You need the quickest way to run a single container with no orchestration. Which service?

A. Azure Container Instances (ACI).

Q3. Which service runs containers that autoscale (even to zero) without you managing Kubernetes?

A. Azure Container Apps.

CHAPTER SUMMARY — KEY TAKEAWAYS

- Containers share the host OS — lighter and faster than VMs.
- ACR stores images; ACI runs a single container fast; Container Apps runs autoscaling serverless containers.
- Size by CPU/memory; Container Apps scales on rules and can scale to zero.

Azure App Service

Chapter 10

What you'll learn

- What App Service and App Service plans are
- Scaling up vs scaling out
- Custom domains, TLS certificates, and networking
- Deployment slots for zero-downtime releases and backups

IN PLAIN ENGLISH

Sometimes you don't want to manage a VM or a container at all — you just have a website or an API's code, and you want to upload it somewhere and have it simply work, with Azure handling the server, the patching, and the plumbing underneath. That's exactly what App Service is for: the closest thing in Azure to "just host my website," with the operating system entirely out of your hands.

10.1 App Service and App Service plans

DEFINITION — Azure App Service

A fully managed platform (PaaS) for hosting web apps, REST APIs, and backends — you deploy your code and Azure handles the servers, patching, and scaling.

Every App Service runs on an App Service plan, which defines the compute (size and tier) and therefore the cost and features. Multiple apps can share one plan and its resources. Tiers range from Free/Shared (dev) up through Basic, Standard, Premium (production features like slots and autoscale).

10.2 Scaling up vs scaling out

Scaling	Means	How
Scale up	Move to a bigger/more powerful plan tier	Change the App Service plan SKU (vertical)
Scale out	Add more instances of the app	Increase instance count, manual or autoscale (horizontal)

MEMORIZE THIS

Scale up = bigger plan (vertical). Scale out = more instances (horizontal, can autoscale on Standard+). The plan tier determines which features (slots, autoscale, custom domains/TLS) are available.

10.3 Custom domains, TLS, networking, backups

- Custom DNS name: map your own domain (verify via DNS records) instead of the default azurewebsites.net.
- TLS/SSL certificates: bind a certificate to secure HTTPS traffic for your custom domain.

- Networking: integrate with a VNet, restrict inbound access, and use private endpoints.
- Backup: configure automatic backups of the app and its content (Standard+).

10.4 Deployment slots

DEFINITION — Deployment slot

A separate, live instance of your app (e.g. a "staging" slot) with its own URL, which you can deploy to and test, then swap into production with no downtime.

ANALOGY

A deployment slot is like rehearsing a play on a second identical stage. When it's perfect, you swap the stages instantly — the audience never sees a gap or a stumble.

MEMORIZE THIS

Deployment slots enable zero-downtime releases via a swap (staging -> production), and instant rollback by swapping back. Slots require the Standard tier or higher.

10.5 Sharpen the edge: plan tiers and swap behavior

Plan tier	What it unlocks
Free / Shared	Dev/test only; shares compute with other customers; no custom domain TLS on Free
Basic	Dedicated compute, custom domains + TLS, manual scale out only (no autoscale)
Standard	Adds deployment slots (5), autoscale, daily backups
Premium	More slots (20), faster hardware, VNet integration, higher scale limits
Isolated	Runs in your own dedicated App Service Environment (ASE) for full network isolation

MEMORIZE THIS

Autoscale specifically requires Standard tier or higher — Basic can scale OUT manually (add instances by hand) but cannot autoscale on a rule/schedule. This is a distinct fact from the slots requirement, and both point to "Standard or higher" for different reasons.

What actually happens during a slot swap

A swap isn't an instant DNS flip — Azure first warms up the target slot by applying its settings and triggering app initialization in the source slot's container, waits for it to respond successfully, and only then redirects traffic. This is what makes it zero-downtime: the app is already warm before it starts receiving production traffic.

EXAM TIP

Slot-specific settings (app settings/connection strings marked "deployment slot setting") stay with the slot and do NOT swap — e.g. staging keeps pointing at a staging database even after a swap, while everything else moves. A question about a swap unexpectedly using the wrong connection string is almost always about this setting.

QUICK CHECK

Cover the answers. Say your answer out loud, then reveal.

Q1. What defines the compute size, tier, and cost your web app runs on?

A. The App Service plan.

Q2. You need to release a new version with no downtime and easy rollback. What do you use?

A. Deployment slots (deploy to staging, then swap).

Q3. Adding more instances of the app is scaling up or scaling out?

A. Scaling out (horizontal).

Q4. What tier is required for deployment slots and autoscale?

A. Standard or higher.

CHAPTER SUMMARY — KEY TAKEAWAYS

- App Service is PaaS web hosting; the App Service plan sets size, tier, cost, and features.
- Scale up = bigger plan; scale out = more instances (autoscale on Standard+).
- Custom domains + TLS secure your app; VNet integration and private endpoints control networking.
- Deployment slots give zero-downtime swaps and instant rollback (Standard+).

Networking

Virtual Networks & Connectivity

Chapter 11 — now includes a dedicated troubleshooting section (11.4), matching the official outline's separate "Troubleshoot network connectivity" bullet under this domain.

What you'll learn

- What a virtual network and subnet are
- Connecting networks with VNet peering
- Public IP addresses and user-defined routes
- Troubleshooting network connectivity

IN PLAIN ENGLISH

Networking is the topic most likely to feel intimidating if you've never worked with IP addresses and routing before — so before anything else, one piece of notation you'll see constantly: 10.0.0.0/16. The first part is a starting IP address. The /16 after the slash says how many addresses belong to that range — a smaller number after the slash means MORE addresses (a /16 gives about 65,000 addresses; a /24 gives 256). You don't need to do the math by hand for this exam, but recognizing that "/16 = a big range, /24 = a smaller range" will make every table in this chapter click.

11.1 Virtual networks and subnets

DEFINITION — Virtual network (VNet)

Your own private, isolated network in Azure, defined by an address space (in CIDR notation, e.g. 10.0.0.0/16), within which resources like VMs communicate securely.

A VNet is subdivided into subnets — smaller address ranges that let you group and isolate resources (for example, a web subnet and a database subnet). Resources in the same VNet can talk to each other by default; subnets let you place security controls between tiers.

ANALOGY

A VNet is a private office building with its own internal phone system. Subnets are the floors — you can put security doors between floors (the web floor and the finance floor) even though they share the same building.

```
# Create a VNet with a subnet
az network vnet create --resource-group myRG --name myVNet \
--address-prefix 10.0.0.0/16 --subnet-name web --subnet-prefix 10.0.1.0/24
```

MEMORIZE THIS

A VNet has an address space; subnets carve it up. Azure reserves 5 IP addresses in every subnet (first 4 + last). Resources in a VNet communicate by default; VNets are isolated from each other unless you connect them.

11.2 VNet peering

To let resources in two separate VNets communicate, you create VNet peering — a low-latency connection over Microsoft's backbone. Peering can be within a region (regional) or across regions (global).

EXAM TIP

Peering is NOT transitive. If VNet A peers with B, and B peers with C, A still can't reach C. You'd need a direct A–C peering, or a hub-and-spoke design with a gateway/firewall routing between them. Peered VNets must not have overlapping address spaces.

11.3 Public IP addresses and user-defined routes

Concept	What to know
Public IP SKU	Basic vs Standard; Standard is secure-by-default (closed until an NSG allows) and zone-redundant
Static vs dynamic	Static keeps the same IP; dynamic can change on deallocation
System routes	Azure's automatic default routing between subnets and to the internet
User-defined route (UDR)	A custom route you add to override defaults — e.g. force traffic through a firewall

MEMORIZE THIS

UDRs override system routes — commonly to force subnet traffic through a network virtual appliance/firewall (next hop). Standard public IPs are closed by default and need an NSG rule to allow inbound. Peering is fast but non-transitive.

11.4 Troubleshooting network connectivity

When a resource can't reach another resource, an endpoint, or the internet, work through the layers in order rather than guessing: is an NSG blocking it (subnet or NIC level)? Is a user-defined route sending traffic somewhere unexpected? Is a peering missing or non-transitive? Is DNS resolving to the address you expect?

Check	What it tells you
NSG rules (subnet + NIC)	Whether a security rule is denying the traffic — use effective security rules (Chapter 12) to see the combined outcome
Effective routes	Whether a UDR is sending traffic to an unexpected next hop instead of the internet or peer VNet
Peering status and transitivity	Whether the two VNets are actually connected, and whether a non-transitive hop is the real cause
DNS resolution	Whether a name is resolving to the public endpoint when a private endpoint was intended, or vice versa

IN PLAIN ENGLISH

Most "why can't X reach Y" scenarios on the exam are really NSG questions or routing questions in disguise. Azure gives you tools that compute the real answer instead of tracing every rule by hand — the portal's Network Watcher blade (covered in depth in Chapter 14) has IP flow verify, next hop, and Connection troubleshoot for exactly this. Use them before assuming a resource is simply broken.

MEMORIZE THIS

For a one-time "is this traffic allowed right now" question, use IP flow verify. For "which route is this packet actually taking," use next hop. For a full end-to-end reachability test between two resources, use Connection troubleshoot. All three live in Network Watcher.

11.5 Sharpen the edge: what those 5 reserved subnet IPs actually are

Every AZ-104 study guide says "5 IPs reserved per subnet" — the exam sometimes expects you to know WHICH five, not just the count. - x.x.x.0 — the network address (identifies the subnet itself) - x.x.x.1 — reserved for the default gateway - x.x.x.2 and x.x.x.3 — reserved to map Azure DNS IPs into the subnet - x.x.x.255 (or the last address in the range) — the broadcast address

MEMORIZE THIS

A /24 subnet has 256 addresses total but only 251 usable once the 5 reserved ones are subtracted — a common "how many usable IPs" calculation question.

User-defined route next-hop types

Next hop type	Sends traffic to
Virtual appliance	A specific IP address, typically a firewall NVA
Virtual network gateway	A VPN/ExpressRoute gateway
VNet peering	Directly to a peered VNet
Internet	The public internet (default for most traffic)
None	Drops the traffic entirely — a deliberate black hole

EXAM TIP

Next hop type "None" is a legitimate, intentional answer when a question describes deliberately blocking a specific traffic pattern at the routing layer rather than with an NSG — don't dismiss it as a distractor.

QUICK CHECK

Cover the answers. Say your answer out loud, then reveal.

Q1. VNet A peers with B, B peers with C. Can A reach C?

A. No — peering isn't transitive; you need direct peering or a hub/gateway.

Q2. How do you force a subnet's outbound traffic through a firewall appliance?

A. A user-defined route (UDR) with the firewall as the next hop.

Q3. How many IP addresses does Azure reserve in each subnet?

A. Five (the first four and the last).

Q4. A VM can't reach another VM. What's the first thing to check?

A. NSG rules (subnet and NIC) and effective routes — the two most common causes.

CHAPTER SUMMARY — KEY TAKEAWAYS

- A VNet is a private network; subnets segment it (Azure reserves 5 IPs per subnet).
- VNet peering connects VNets over the backbone but is non-transitive; no overlapping address spaces.
- UDRs override default routing (e.g. via a firewall); Standard public IPs are closed by default.
- Troubleshoot connectivity by checking NSGs, effective routes, peering, and DNS, in that order — Network Watcher automates each check.

Securing Access to Virtual Networks

Chapter 12

What you'll learn

- Filtering traffic with network security groups (NSGs)
- Simplifying rules with application security groups (ASGs)
- Reaching PaaS privately with service and private endpoints
- Secure VM access with Azure Bastion

IN PLAIN ENGLISH

Building a network in Chapter 11 was like constructing a building. This chapter is about deciding who gets to walk through which doors. Think of everything here as answering one question over and over, in different contexts: "should THIS traffic, from THIS source, be allowed to reach THAT destination?" NSGs are the main tool for answering it, and the rest of the chapter covers more specialized doors (for admin access, for reaching Microsoft's own services privately).

12.1 Network security groups (NSGs)

DEFINITION — Network security group (NSG)

A set of inbound and outbound security rules that allow or deny network traffic to subnets and network interfaces, based on source, destination, port, and protocol.

An NSG is your basic packet filter. Each rule has a priority number; rules are evaluated from lowest number to highest, and the first matching rule wins — later rules are ignored for that traffic. NSGs come with default rules you can't delete (but can override).

Rule element	Detail
Priority	100–4096; lower number = evaluated first; first match wins
Direction	Separate inbound and outbound rule sets
Default inbound	Deny from internet (allow within VNet and from load balancer)
Default outbound	Allow to internet and within VNet

MEMORIZE THIS

NSG rules: priority 100–4096, lowest first, first match wins. By default inbound internet is denied and outbound internet is allowed. An NSG can be attached to a subnet and/or a NIC.

Evaluating effective security rules

A single VM's traffic can be filtered by two NSGs at once — one attached to its subnet and one attached to its NIC — plus Azure's own default rules. Traffic must be allowed by both NSGs to get through. Rather than manually combining every rule set in your head, Azure computes this for you.

DEFINITION — Effective security rules

The final, combined set of NSG rules that actually apply to a specific network interface, after merging the NIC-level NSG, the subnet-level NSG, and Azure's default rules — viewable per NIC in the portal or via Network Watcher.

```
# View effective security rules for a NIC (CLI)
az network nic list-effective-nsg --name myNIC --resource-group myRG
```

ANALOGY

If a subnet NSG allows a port but the NIC's own NSG denies it, the traffic is still blocked — like needing two separate keys to unlock one door. "Effective rules" shows you the result after both locks are considered, so you don't have to trace it by hand.

MEMORIZE THIS

Traffic to a VM must pass both the subnet NSG and the NIC NSG (if both exist) — the more restrictive result wins. Use effective security rules (portal: NIC -> Networking -> Effective security rules, or `az network nic list-effective-nsg`) to see the real, combined outcome instead of guessing.

Sharpen the edge: the default rules by exact name and priority

Default rule	Priority	Direction	Effect
AllowVNetInBound	65000	Inbound	Allow all traffic from the VNet (incl. peered VNets)
AllowAzureLoadBalancerInBound	65001	Inbound	Allow Azure Load Balancer health probes
DenyAllInBound	65500	Inbound	Deny everything else
AllowVNetOutBound	65000	Outbound	Allow all traffic to the VNet
AllowInternetOutBound	65001	Outbound	Allow outbound traffic to the internet
DenyAllOutBound	65500	Outbound	Deny everything else

MEMORIZE THIS

These six default rules sit at priorities 65000–65500 — outside your custom range of 100–4096 — so any rule you create automatically outranks them. Note the trap in the name: `AllowInternetOutBound` is a DEFAULT rule, meaning every VM can reach the internet unless you explicitly override it with a lower-priority deny.

EXAM TIP

An NSG allows up to 1,000 rules per NSG, and each source/destination field normally holds one prefix or IP — but augmented security rules let a single rule hold up to 4,000 IP addresses/ranges or ports in one field, collapsing what would otherwise be hundreds of near-duplicate rules into one.

12.2 Application security groups (ASGs)

Writing NSG rules by IP address is brittle. Application security groups let you group VMs by role (say, "web" or "db") and then write NSG rules that reference the group name instead of IPs — so as VMs come and go, the rules still apply.

ANALOGY

Instead of a guest list naming every individual (IP addresses), an ASG is a category — "band members" — and the rule says "band members may enter." New members are covered automatically.

12.3 Service endpoints and private endpoints

By default, Azure PaaS services (like a storage account) have public endpoints. Two features let your VNet reach them more privately:

Feature	What it does	Security level
Service endpoint	Routes VNet traffic to the PaaS service over the Azure backbone; the service still has a public IP but can be restricted to your VNet	Good
Private endpoint	Gives the PaaS service a private IP inside your VNet; traffic never touches the public internet	Best / most secure

MEMORIZE THIS

Service endpoint = optimized/secured route to a PaaS service (still a public IP, restricted to your VNet).
Private endpoint = a private IP for the service inside your VNet (most secure; the usual "no public exposure" answer).

12.4 Azure Bastion

To administer a VM you normally need RDP/SSH, which traditionally means giving the VM a public IP — a security risk. Azure Bastion is a managed service that provides secure RDP/SSH to your VMs straight from the Azure portal over TLS, so the VMs need no public IP and no open management ports to the internet.

MEMORIZE THIS

Azure Bastion = browser-based RDP/SSH with no public IP on the VM and no exposed 3389/22 to the internet. Deployed into a dedicated AzureBastionSubnet.

12.5 Sharpen the edge: subnet sizing and private endpoint DNS

- The AzureBastionSubnet must be named exactly that (case-sensitive) and Microsoft recommends a /26 prefix at minimum — a question testing a failed Bastion deployment may hinge on the subnet being too small or misnamed.
- A private endpoint for a PaaS service needs a corresponding private DNS zone linked to your VNet so that the service's normal public hostname (e.g. mystorage.blob.core.windows.net) resolves to the new private IP instead of the public one — without that DNS piece, name resolution still points at the public endpoint even though the private endpoint exists.
- A service endpoint requires no DNS changes at all — it works by adding your VNet's identity to the request over the existing public hostname, which is part of why it's simpler to set up (but less isolated) than a private endpoint.

EXAM TIP

"We created a private endpoint but traffic is still going out over the public internet" is a very common scenario question. The almost-always-correct answer is a missing or unlinked private DNS zone — the private endpoint exists and works, but nothing is telling clients to resolve to it.

QUICK CHECK

Cover the answers. Say your answer out loud, then reveal.

Q1. Two NSG rules match the same traffic. Which applies?

A. The one with the lower priority number (first match wins).

Q2. A VM has an NSG on its subnet AND on its NIC. Which one filters the traffic?

A. Both — traffic must be allowed by each; view the combined outcome as effective security rules.

Q3. How do you write NSG rules that keep working as web VMs are added/removed?

A. Use an application security group (ASG) and reference it in the rule.

Q4. Which gives a PaaS service a private IP in your VNet with no public exposure?

A. A private endpoint.

Q5. How do you RDP to a VM without giving it a public IP?

A. Use Azure Bastion.

CHAPTER SUMMARY — KEY TAKEAWAYS

- NSGs allow/deny traffic by priority (100–4096, lowest first, first match wins); default denies inbound internet.
- Effective security rules merge subnet + NIC NSGs into the real, combined outcome — view them instead of guessing.
- ASGs group VMs by role so rules survive IP changes.
- Service endpoints secure the route to PaaS; private endpoints give it a private IP (most secure).
- Azure Bastion provides RDP/SSH with no public IP on the VM.

Name Resolution & Load Balancing

Chapter 13

What you'll learn

- Resolving names with Azure DNS and private DNS zones
- Distributing traffic with Azure Load Balancer
- Choosing between load-balancing options
- Troubleshooting load balancing

IN PLAIN ENGLISH

This chapter answers two very ordinary questions that just happen to need special tools at cloud scale. First: "when someone types a website name, how does that turn into the right server?" (that's DNS — nothing more mysterious than a phone book matching names to numbers). Second: "if I have five identical servers instead of one, how do visitors get spread across all of them instead of hammering just one?" (that's load balancing). Neither idea is new to computing — Azure just gives you managed services to do both without running the infrastructure yourself.

13.1 Azure DNS

DNS translates names (like app.contoso.com) into IP addresses. Azure DNS hosts your public domains on Azure's name servers, while private DNS zones provide name resolution inside your VNets — including automatic registration of VM names — without exposing anything publicly.

MEMORIZE THIS

Azure DNS hosts public domains. Private DNS zones resolve names within your VNets and can auto-register VM records. DNS doesn't move traffic — it just returns addresses.

13.2 Azure Load Balancer

DEFINITION — Azure Load Balancer

A Layer-4 (TCP/UDP) load balancer that distributes incoming traffic across a backend pool of VMs for high availability and scale.

A load balancer spreads connections across healthy backend VMs so no single VM is overwhelmed and a failed VM is bypassed. It can be public (internet-facing) or internal (private, for traffic inside your network). Health probes decide which backends are healthy.

Term	Meaning
Frontend IP	Where traffic arrives (public or private)
Backend pool	The VMs that receive the traffic

Term	Meaning
Health probe	Checks backend health; unhealthy ones are removed from rotation
Load-balancing rule	Maps a frontend port to a backend port

EXAM TIP

The exam contrasts load-balancing options. Azure Load Balancer = Layer 4 (TCP/UDP, regional). Application Gateway = Layer 7 (HTTP/S, URL-based routing, includes WAF). Front Door / Traffic Manager = global distribution. Pick by layer and scope.

13.3 Troubleshooting load balancing

When traffic won't flow through a load balancer specifically, check: is the load balancer's health probe passing (an unhealthy probe removes the backend)? Are the load-balancing rules mapping the right ports? For general connectivity issues upstream of the load balancer (NSGs, routes), see the troubleshooting steps in Chapter 11 — Network Watcher tools like IP flow verify and connection troubleshoot help pinpoint where traffic is blocked either way.

MEMORIZE THIS

For load-balancer problems, a failing health probe is the classic cause — the backend gets pulled from rotation. For general connectivity, check NSG rules and effective routes first (Chapter 11).

13.4 Sharpen the edge: Load Balancer SKUs and Traffic Manager routing

	Basic Load Balancer	Standard Load Balancer
Backend pool	Availability set or single VMs	Any VM, VMSS, or Availability Zone-spanning setup
Availability Zone support	No	Yes — zone-redundant or zonal frontend
Security	Open by default	Secure by default — requires an explicit NSG to allow traffic
SLA	None	99.99%

MEMORIZE THIS

Standard Load Balancer is secure-by-default (needs an NSG to allow traffic through) while Basic is open — exactly the opposite pattern from what some people assume, and a common trap. Standard is also required for any zone-redundant design.

Traffic Manager routing methods

Method	Routes based on
Priority	A primary endpoint, with others as failover order

Method	Routes based on
Weighted	A distribution ratio you assign across endpoints
Performance	Lowest network latency to the user
Geographic	The user's geographic location, for data-residency/compliance routing
Multivalued	Returns several healthy endpoint IPs at once (for IPv4/IPv6 endpoints)
Subnet	The user's source IP range, mapped to a specific endpoint

EXAM TIP

Geographic routing is the one to reach for when a question mentions data residency or "users in the EU must be served from EU endpoints only" — Performance routing optimizes speed, not compliance, and is a common wrong answer for residency scenarios.

QUICK CHECK

Cover the answers. Say your answer out loud, then reveal.

Q1. Which service resolves names privately inside your VNets and can auto-register VMs?

A. Private DNS zones.

Q2. At which layer does Azure Load Balancer operate?

A. Layer 4 (TCP/UDP).

Q3. You need URL-based routing and a web application firewall. Which service?

A. Application Gateway (Layer 7).

Q4. VMs behind a load balancer get no traffic. What's the classic first thing to check?

A. The health probe (a failing probe removes backends from rotation).

CHAPTER SUMMARY — KEY TAKEAWAYS

- Azure DNS hosts public domains; private DNS zones resolve names inside VNets.
- Azure Load Balancer is Layer 4; Application Gateway is Layer 7 + WAF; Front Door/Traffic Manager are global.
- Load balancers use frontend IP, backend pool, health probes, and rules; failing probes drop backends.

Monitoring & Maintenance

Monitoring with Azure Monitor

Chapter 14 — reordered to follow the official bullet sequence: metrics/logs and log settings, then querying logs, then alerts, then Insights, then Network Watcher.

What you'll learn

- The difference between metrics and logs, and how to turn on log collection
- Querying logs in Log Analytics with KQL
- Alerting with alert rules, action groups, and processing rules
- Pre-built dashboards with Azure Monitor Insights
- Diagnosing networks with Network Watcher

IN PLAIN ENGLISH

Once resources are up and running, a new question replaces "how do I build this?" with "is everything actually okay, and how would I know if it wasn't?" This chapter is entirely about visibility — the dashboards, logs, and automatic notifications that tell you a disk is filling up, a VM is overloaded, or something has gone quietly wrong, ideally before a user ever notices.

14.1 Azure Monitor: metrics vs logs

Azure Monitor is the umbrella for observing everything in Azure. It collects two fundamentally different kinds of data, and telling them apart is a reliable exam point.

	Metrics	Logs
What they are	Numeric values sampled over time (CPU %, disk IOPS)	Detailed, timestamped event records
Speed	Lightweight, near real-time	Richer but heavier to query
Stored in	The metrics database	A Log Analytics workspace
Queried with	Metrics explorer / charts	KQL (Kusto Query Language)

ANALOGY

Metrics are the speedometer — a constant numeric readout, great for instant alerts. Logs are the trip diary — detailed entries you search later to reconstruct what happened.

MEMORIZE THIS

Metrics = numeric, near-real-time, ideal for fast alerting. Logs = rich event records in a Log Analytics workspace, queried with KQL.

Configuring log settings (diagnostic settings)

Most Azure resources don't send their logs anywhere by default — you have to turn it on. A diagnostic setting on a resource picks which log categories and metrics to export, and where: a Log Analytics workspace (for

querying), a storage account (for cheap long-term archiving), or an event hub (for streaming to another system). You can send to more than one destination at once.

```
# Send a VM's logs to a Log Analytics workspace (CLI)
az monitor diagnostic-settings create --name sendToLAW \
--resource myVM --resource-group myRG \
--workspace myWorkspace --logs '[{"category":"Administrative","enabled":true}]'
```

MEMORIZE THIS

Diagnostic settings are what turn on log collection for a resource and choose the destination(s). No diagnostic setting = no logs flowing anywhere, even though metrics are collected automatically by default.

14.2 Querying and analyzing logs with KQL

Once logs land in a Log Analytics workspace, you query them with KQL (Kusto Query Language) — a readable, pipe-based syntax. A basic query starts with a table name and filters/shapes the data with operators chained by |.

```
# Find VM shutdown events in the last 24 hours
AzureActivity
| where TimeGenerated > ago(24h)
| where OperationNameValue == "MICROSOFT.COMPUTE/VIRTUALMACHINES/DEALLOCATE/ACTION"
| project TimeGenerated, Caller, ResourceGroup
| order by TimeGenerated desc
```

IN PLAIN ENGLISH

Read it top to bottom: start from a table, keep only rows matching where, pick the columns you want with project, and sort with order by. Each line narrows or reshapes the previous result.

14.3 Alerts, action groups, and processing rules

Monitoring is only useful if it tells someone. Azure Monitor alert rules fire when a condition is met (a metric threshold or a log query result). What happens next is defined by an action group. A separate alert processing rule can modify alerts in bulk (e.g. suppress during maintenance).

Component	Role
Alert rule	The condition that triggers an alert (metric, log, or activity-log based)
Action group	The set of actions taken when it fires: email/SMS, webhook, Logic App, Automation runbook
Alert processing rule	Applies actions or suppression to alerts in bulk (e.g. mute during a maintenance window)

MEMORIZE THIS

Alert rule = the trigger condition; action group = who/what responds (notifications + automated actions). Alert processing rules suppress or modify alerts in bulk, e.g. during planned maintenance.

14.4 Azure Monitor Insights

Rather than building dashboards from scratch, Azure Monitor ships pre-built, curated monitoring experiences called Insights for the resource types you'll manage most as an administrator.

Insight	What it shows
VM insights	Performance (CPU/memory/disk/network) and a map of processes and dependencies between VMs
Storage account insights	Capacity, availability, and performance/transaction metrics for a storage account
Network insights	Topology and health across VNets, NSGs, load balancers, gateways, and connections
Application Insights	Application performance monitoring — requests, failures, dependencies, response times

MEMORIZE THIS

Insights = pre-built dashboards layered on top of the same metrics/logs you already collect — you don't have to build the views yourself. VM insights needs the Azure Monitor Agent; Application Insights is for app-level (not infrastructure-level) telemetry.

14.5 Network Watcher

Network Watcher is the toolbox for diagnosing network problems. Key tools: IP flow verify (is an NSG allowing this traffic?), next hop (where will this packet go?), connection troubleshoot / Connection Monitor (test and continuously monitor connectivity), and NSG diagnostics.

MEMORIZE THIS

Network Watcher diagnoses connectivity: IP flow verify checks if an NSG allows/denies traffic; next hop shows routing; Connection Monitor continuously watches connectivity between endpoints.

14.6 Sharpen the edge: retention defaults and diagnostic precision

Data type	Default retention	Configurable range
Platform metrics	93 days	Not extendable in the metrics store itself — route to Log Analytics for longer
Log Analytics workspace logs	30 days (free/basic)	Up to 730 days (2 years) per table, at added cost
Activity log	90 days (kept automatically)	Export to a Log Analytics workspace or storage account for longer

MEMORIZE THIS

If a scenario needs log data kept longer than the default window, the answer is almost always "send it to a Log Analytics workspace (or storage account for cheap long-term archive) via diagnostic settings" — not "wait, isn't it kept forever already?" Nothing is retained indefinitely without explicit configuration.

IP flow verify vs NSG diagnostics vs Connection troubleshoot

Tool	Answers
IP flow verify	"Is this specific traffic allowed or denied, and by which rule?" — a single point-in-time check
NSG diagnostics	"Given this traffic, which NSG and rule decided the outcome, across a chain of NSGs?" — broader than IP flow verify
Connection troubleshoot	"Can VM A actually reach VM B/a URL right now?" — tests the full path end to end, including latency
Connection Monitor	The same test as above, but running continuously over time with alerting

EXAM TIP

A one-time "why is this blocked right now" question points to IP flow verify. An ongoing "alert us if connectivity between these two sites degrades" requirement points to Connection Monitor, not a single troubleshoot run.

QUICK CHECK

Cover the answers. Say your answer out loud, then reveal.

Q1. Which data type is best for a fast alert on high CPU: metrics or logs?

A. Metrics (numeric, near-real-time).

Q2. A resource has no diagnostic setting configured. Are its logs being collected anywhere?

A. No — without a diagnostic setting, logs aren't exported to a workspace, storage, or event hub.

Q3. What language queries logs in a Log Analytics workspace, and what does 'where' do in it?

A. KQL; 'where' filters rows by a condition.

Q4. An alert fires — what defines whether it emails you or runs a runbook?

A. The action group.

Q5. Which Insight would you check for CPU/memory trends and process dependencies across VMs?

A. VM insights.

Q6. Which Network Watcher tool tells you if an NSG is blocking specific traffic?

A. IP flow verify.

CHAPTER SUMMARY — KEY TAKEAWAYS

- Azure Monitor collects metrics (numeric, fast) and logs (rich, KQL in Log Analytics); diagnostic settings turn on log collection and choose the destination(s) — nothing flows without one.
- KQL queries logs with chained operators like where/project/order by.
- Alert rules trigger; action groups respond; alert processing rules modify/suppress alerts in bulk.
- Insights (VM/Storage/Network/Application) give pre-built dashboards on top of the same data.
- Network Watcher (IP flow verify, next hop, Connection Monitor) diagnoses network issues.

Backup & Recovery

Chapter 15

What you'll learn

- Protecting data with Azure Backup, Recovery Services vaults, and Azure Backup vaults
- Creating and applying backup policies, and monitoring backups with reports and alerts
- Replicating and failing over workloads with Azure Site Recovery
- The crucial difference between backup and disaster recovery

IN PLAIN ENGLISH

Everything in this book so far has assumed things go right. This last chapter is about what happens when they don't — someone deletes the wrong file, ransomware encrypts a server, or an entire Azure region goes offline. There are really only two questions to answer here: "how do I get a copy of my data back?" (backup) and "how do I keep the whole service running somewhere else if the primary location is gone?" (disaster recovery). They sound similar but solve different problems, and mixing them up is one of the most common real-world mistakes this exam checks for.

15.1 Azure Backup and vaults

DEFINITION — Azure Backup

A service that backs up data — VMs, files/folders, Azure Files, SQL in VMs — into a vault, so you can restore to a previous point in time after loss, corruption, or ransomware.

Backups are stored in a vault, and there are now two vault types — know when each applies.

Vault type	Backs up	Notes
Recovery Services vault	VMs, SQL/SAP HANA in Azure VMs, Azure Files (via agent), on-prem via MARS agent	The original, broadest vault; also used by Azure Site Recovery
Azure Backup vault	Newer workloads: Azure Database for PostgreSQL, Blob storage (operational backup), disks	Purpose-built for newer, cloud-native data sources; simpler vault-level management

A backup policy defines the schedule (how often) and retention (how long to keep recovery points), and is attached to the item you're protecting.

```
# Enable backup for a VM into a Recovery Services vault (portal or CLI)
az backup protection enable-for-vm --resource-group myRG \
--vault-name myVault --vm myVM --policy-name DefaultPolicy
```

MEMORIZE THIS

Azure Backup creates restorable recovery points. Use a Recovery Services vault for VMs/SQL/on-prem agents (and Site Recovery); use an Azure Backup vault for newer workloads like PostgreSQL and Blob operational backup. A backup policy = schedule + retention.

Reports and alerts for backups

Protecting data is only half the job — you also need to know when a backup fails or a job is behaving unusually. Azure Backup provides both built-in monitoring and a reporting workbook.

Capability	What it does
Backup alerts (built-in)	Automatically fire on backup failures, no valid policy, or expiring soft-deleted items — visible in the vault and Azure Monitor alerts
Backup Reports	A Power BI-style workbook (backed by a Log Analytics workspace) summarizing backup jobs, storage consumed, and compliance across many vaults over time
Email notifications	Configure the vault to email on job failures or all events

MEMORIZE THIS

Backup alerts are built-in and fire automatically on failures. Backup Reports need a Log Analytics workspace connected to the vault and give a long-term, cross-vault view of jobs, consumption, and compliance — useful for proving backups are actually happening, not just configured.

15.2 Azure Site Recovery (ASR)

DEFINITION — Azure Site Recovery (ASR)

A disaster-recovery service that continuously replicates whole workloads (VMs) to another region, so you can fail over and keep running if the primary region goes down.

ANALOGY

Backup is a photo album — snapshots you restore from after something is lost. Site Recovery is a stunt double standing by in another city — if the star (your region) goes down mid-show, the double steps in and the show continues.

MEMORIZE THIS

Azure Backup = restore data from points in time. Azure Site Recovery = replicate and fail over whole workloads to another region for business continuity. Different jobs — don't confuse them.

15.3 Backup vs disaster recovery — the key distinction

	Azure Backup	Azure Site Recovery
Purpose	Recover lost/corrupted data	Keep running through a regional disaster
What it produces	Recovery points to restore from	A replicated copy you fail over to
Answers	"Get my data back"	"Keep my service online"

EXAM TIP

If a question is about recovering deleted/corrupted files or a VM to an earlier state -> Azure Backup. If it's about surviving a whole region outage with minimal downtime -> Azure Site Recovery (failover to a secondary region).

15.4 Sharpen the edge: backup policy structure and RPO/RTO

A backup policy isn't just "daily or weekly" — it's a layered retention schedule, and the exam expects you to read one correctly.

Policy layer	Typical example
Backup frequency	Once daily (VM backup) or more often for some workloads (e.g. every few hours for SQL)
Retention: daily points	Keep each daily recovery point for, say, 30 days
Retention: weekly points	Keep one recovery point per week for, say, 12 weeks
Retention: monthly points	Keep one per month for, say, 12 months
Retention: yearly points	Keep one per year for, say, 5–10 years

MEMORIZE THIS

These layers stack — a single recovery point can simultaneously count as "this week's weekly point" and "this month's monthly point," so you don't need separate backup jobs for each retention tier. This is how Azure Backup keeps years of history without storing a full daily copy forever.

RPO and RTO — the vocabulary behind "why do we even need this" questions

Term	Means	Which Azure service primarily addresses it
RPO (Recovery Point Objective)	How much data loss (in time) is acceptable — "we can lose up to 1 hour of data"	Backup frequency (Backup) / replication frequency (Site Recovery)
RTO (Recovery Time Objective)	How long recovery is allowed to take — "we must be back up within 4 hours"	Site Recovery's failover speed; Backup's restore speed

EXAM TIP

A scenario stating a maximum acceptable data loss window is describing an RPO requirement — tighten it by backing up/replicating more frequently. A scenario stating a maximum acceptable downtime window is an RTO requirement — tighten it with Site Recovery and a tested failover plan, not just more frequent backups.

QUICK CHECK

Cover the answers. Say your answer out loud, then reveal.

Q1. Where are VM backups stored, and what defines their schedule and retention?

A. A Recovery Services vault; a backup policy.

Q2. You need to back up an Azure Database for PostgreSQL instance. Which vault type?

A. An Azure Backup vault (purpose-built for newer cloud-native workloads).

Q3. How would you find out a backup job failed last night without checking manually?

A. A backup alert (built-in, fires automatically on failure).

Q4. How do you get a long-term, cross-vault view of backup compliance and storage consumed?

A. Backup Reports, via a Log Analytics workspace connected to the vault(s).

Q5. Ransomware encrypted a VM's data. Which service restores it?

A. Azure Backup (restore from a clean recovery point).

Q6. An entire Azure region goes offline. Which service keeps your workload running elsewhere?

A. Azure Site Recovery (fail over to the secondary region).

Q7. In one line: backup vs Site Recovery?

A. Backup restores data from points in time; Site Recovery replicates and fails over workloads to another region.

CHAPTER SUMMARY — KEY TAKEAWAYS

- Azure Backup creates restorable recovery points; policy = schedule + retention.
- Recovery Services vault = VMs/SQL/on-prem/Site Recovery; Azure Backup vault = newer workloads (PostgreSQL, Blob).
- Backup alerts fire automatically on failures; Backup Reports (via Log Analytics) give long-term, cross-vault visibility.
- Azure Site Recovery replicates workloads to another region and fails over for continuity.
- Backup = get data back; Site Recovery = keep the service running through a disaster.

Appendix

Appendix A: Glossary

Quick definitions for the terms AZ-104 expects you to know. A final-night sweep.

Term	Meaning
Management group	Container for many subscriptions; applies governance at scale.
Subscription	Billing and access boundary for resources.
Resource group	Folder holding related resources with a shared lifecycle.
Azure Resource Manager (ARM)	The management layer every request passes through.
Region / availability zone	Geographic datacenter set / a separate datacenter within a region.
Member vs guest user	Internal account vs external B2B collaborator (#EXT#).
Dynamic group	Membership set by an attribute rule; needs Entra ID P1.
SSPR	Self-service password reset (register methods first).
Azure RBAC	Access to Azure resources via role assignments (principal+role+scope).
Owner / Contributor / Reader	Full+grant / manage-but-not-grant / view-only.
User Access Administrator	Manages role assignments only.
Entra role	Directory admin role (e.g. Global Administrator) — not Azure RBAC.
Azure Policy	Enforces/audits what resources are allowed (Audit/Deny/DINE).
Initiative	A group of Azure policies managed together.
Resource lock	CanNotDelete or ReadOnly; inherited; overrides RBAC for delete.
Tag	Key/value label for organizing and cost reporting (not inherited).
Budget / Advisor	Spend threshold with alerts / recommendations across 5 pillars.
Storage account	Container for Blob/Files/Queue/Table services.

Term	Meaning
LRS/ZRS/GRS/GZRS	Local / zone / geo / zone+geo redundancy; RA- adds read secondary.
Access key	Full-access account key (two, for rotation).
SAS	Scoped, time-limited signed URL; use stored access policy to revoke.
Access tier	Hot/Cool/Cold/Archive; Archive is offline (rehydrate).
Soft delete / versioning / snapshot	Recover deleted / prior versions / point-in-time copies.
Lifecycle management	Auto-tier or delete blobs by age/last access.
ARM template / Bicep	JSON IaC / friendlier IaC that transpiles to ARM; both idempotent.
VM size (SKU)	Sets CPU/RAM; resize when stopped.
Availability Set	Fault + update domains within a datacenter (~99.95%).
Availability Zones	Separate datacenters within a region (~99.99%).
VMSS	Virtual Machine Scale Set — identical auto-scaling VMs.
Scale up / out	Bigger instance (vertical) / more instances (horizontal).
ACR / ACI / Container Apps	Image registry / single container / autoscaling serverless containers.
App Service / plan	PaaS web hosting / defines size, tier, cost, features.
Deployment slot	Staging instance swapped to prod for zero-downtime (Standard+).
VNet / subnet	Private network / segment of it (Azure reserves 5 IPs/subnet).
VNet peering	Connects VNets over the backbone; non-transitive.
UDR	User-defined route; overrides default routing (e.g. via firewall).
NSG	Allow/deny rules; priority 100–4096, lowest first, first match wins.
ASG	Application security group; groups VMs by role for NSG rules.

Term	Meaning
Service / private endpoint	Secured route to PaaS / private IP for PaaS (most secure).
Azure Bastion	RDP/SSH to VMs with no public IP.
Azure DNS / private DNS zone	Host public domains / resolve names inside VNets.
Azure Load Balancer	Layer-4 (TCP/UDP) traffic distribution; health probes.
Application Gateway	Layer-7 (HTTP) routing + WAF.
Azure Monitor	Umbrella for metrics, logs, alerts, insights.
Metrics vs logs	Numeric near-real-time / rich records queried with KQL.
Alert rule / action group	Trigger condition / the response actions.
Network Watcher	Network diagnostics (IP flow verify, next hop, Connection Monitor).
Azure Backup / Recovery Services vault	Restore data from recovery points / where backups live.
Azure Site Recovery	Replicate and fail over workloads to another region.
RPO / RTO	Max acceptable data loss (time) / max acceptable downtime (time).
NotActions	Subtracts permissions from a custom role's Actions — NOT a Deny; another role assignment can still grant them.
Complete vs Incremental deployment mode	Deletes resources not in the template / leaves extras alone (default).
Append vs Modify (policy)	Adds a value at creation only / can also update existing resources.
User delegation SAS	Signed with Entra ID credentials, not the account key — the most secure SAS type.
AllowVNetInBound / DenyAllInBound	Default NSG rules at priority 65000 / 65500; outside your 100–4096 custom range.
Augmented security rules	One NSG rule holding up to 4,000 IPs/ranges, collapsing many near-duplicate rules.
Slot-specific setting	An app setting/connection string that stays with a slot and does NOT move during a swap.
Rehydration priority	Standard (~15h) vs High (under 1h, best-effort) for reading an Archive-tier blob.

Appendix B: Exam-Day Strategy

- Do the labs early. AZ-104 has interactive lab tasks in a live portal. Practicing real clicks beats memorizing screenshots. If labs come first, budget your time so you don't run short.
- Read the last sentence first in long scenario questions to know what's actually asked.
- Watch the near-twins: Backup vs Site Recovery; Availability Set vs Zones; service vs private endpoint; scale up vs out; Azure RBAC vs Entra roles; Contributor vs Owner; NSG vs Azure Firewall.
- Recognize both syntaxes: Azure CLI (az ...) and PowerShell (Verb-AzNoun). Questions show either.
- Remember the constants: NSG priority lowest-first; Archive is offline; peering non-transitive; Contributor can't assign roles.
- Flag and move on. ~100 minutes for 40–60 items plus labs — don't stall. Answer everything; no penalty for guessing.
- Trust the green boxes. The MEMORIZE facts are the exact distinctions AZ-104 rewards.

EXAM TIP

Aim for 85%+ on Microsoft's free official AZ-104 practice assessment, and be able to perform each task once by hand in a real subscription, before you book.

Appendix C: CLI & PowerShell Quick Reference

The most common verbs, both ways. Recognize the pattern: CLI is az <service> <action>; PowerShell is <Verb>-Az<Noun>.

Task	Azure CLI	PowerShell
Create resource group	az group create	New-AzResourceGroup
Create VM	az vm create	New-AzVM
List VMs	az vm list	Get-AzVM
Create storage account	az storage account create	New-AzStorageAccount
Create VNet	az network vnet create	New-AzVirtualNetwork
Create NSG	az network nsg create	New-AzNetworkSecurityGroup
Assign a role	az role assignment create	New-AzRoleAssignment
Deploy template	az deployment group create	New-AzResourceGroupDeployment
Enable VM backup	az backup protection enable-for-vm	Enable-AzRecoveryServicesBackupProtection

MEMORIZE THIS

CLI verbs: create / list / show / delete / update. PowerShell verbs: New- / Get- / Set- / Remove-. If you see az it's CLI; if you see -Az it's PowerShell.

Appendix D: Practice Question Bank

Forty exam-style questions across all five parts — the last ten target the precise numbers and traps from each chapter's "Sharpen the Edge" section. Cover the answer key and test yourself.

1. List the Azure management hierarchy from broadest to narrowest. 2. Which built-in role can manage all resources but cannot assign roles to others? 3. You assign Reader at the subscription scope. What does the user see? 4. A group's membership should update automatically as people join Sales. What do you configure (and what license)? 5. Which governance tool blocks creation of resources that violate a rule? 6. Which resource lock allows changes but prevents deletion? 7. Does a budget stop resources when exceeded? 8. Which redundancy option is the cheapest that survives a full regional outage? 9. Is storage encryption at rest something you must enable? 10. You must grant a partner read-only access to one container for 24 hours. What do you issue? 11. What lets you revoke many SAS tokens centrally? 12. Most secure way for an app to access blobs with no stored secret? 13. Which access tier is offline and must be rehydrated before reading? 14. How do you automatically move 180-day-old blobs to Archive? 15. What does Bicep compile to, and are deployments idempotent? 16. Which protects VMs against an entire datacenter failure: availability set or zones? 17. What does an availability set spread VMs across? 18. Which service auto-scales identical VM instances? 19. Fastest way to run a single container with no orchestration? 20. Which service runs autoscaling serverless containers without managing Kubernetes? 21. What App Service feature gives zero-downtime releases, and what tier is required? 22. VNet A peers B, B peers C. Can A reach C? 23. How do you force a subnet's traffic through a firewall appliance? 24. Two NSG rules match the same traffic — which wins? 25. Which gives a PaaS service a private IP inside your VNet? 26. How do you RDP to a VM without a public IP? 27. At which layer does Azure Load Balancer operate, and Application Gateway? 28. Metrics or logs for a fast CPU alert? What queries logs? 29. An alert fires — what defines the notification/automation response? 30. Region-wide outage: which service keeps the workload running elsewhere? 31. A single VM has a Premium SSD OS disk but a Standard SSD data disk. What SLA applies? 32. Can you add an already-running VM to an availability set after the fact? 33. A custom role's NotActions excludes an action. Can a user still get that permission? 34. You deploy an ARM template in Complete mode. What happens to resources in the RG that aren't in the template? 35. Which SAS type is signed with Entra ID credentials instead of the storage account key? 36. At what priority do the default NSG rules sit, and why does that matter for custom rules? 37. A staging slot points at a staging database. After a swap to production, which database does it use? 38. Rehydrating an Archive blob for an urgent same-hour request — which rehydration priority, and is the timing guaranteed? 39. Which Traffic Manager routing method fits a requirement that EU users must be served only from EU endpoints? 40. A scenario states 'we can tolerate at most 1 hour of data loss.' Is that describing RPO or RTO?

Answer Key

1. Management group -> subscription -> resource group -> resource. 2. Contributor. 3. Every resource group and resource in that subscription (inheritance). 4. A dynamic group with an attribute rule; Entra ID P1. 5. Azure Policy (Deny effect). 6. CanNotDelete. 7. No — it triggers alerts only. 8. GRS (geo-redundant). 9. No — it's on by default (Microsoft-managed keys). 10. A SAS scoped to that container, read-only, 24-hour expiry. 11. A stored access policy the SAS references. 12. A managed identity with Entra ID + RBAC. 13. Archive. 14. A blob lifecycle management rule. 15. ARM JSON; yes, idempotent. 16. Availability Zones. 17. Fault domains and update domains (within a datacenter). 18. A Virtual Machine Scale Set (VMSS). 19. Azure Container Instances (ACI). 20. Azure Container Apps. 21. Deployment slots (swap); Standard tier or higher. 22. No — peering is non-transitive. 23. A user-defined route (UDR) with the firewall as next hop. 24. The rule with the lower priority number (first match wins). 25. A private endpoint. 26. Azure Bastion. 27. Load Balancer = Layer 4; Application Gateway = Layer 7 (+WAF). 28. Metrics for the fast alert; KQL queries logs (in Log Analytics). 29. The action group. 30. Azure Site Recovery (failover to secondary region). 31. No 99.9% guarantee —

EVERY disk (OS and all data disks) must be Premium SSD/Ultra Disk for the single-VM 99.9% SLA to apply. 32. No — an availability set can only be chosen at VM creation time. 33. Yes — NotActions only subtracts from that specific role definition; a different role assignment can still grant it (it's not a Deny). 34. Any resource in the resource group that ISN'T in the template gets deleted. 35. A user delegation SAS. 36. 65000–65500 — outside your custom 100–4096 range, so any custom rule you create automatically outranks the defaults. 37. The staging database — slot-specific settings stay with the slot and do not move during a swap. 38. High priority; no, it's best-effort with no guaranteed SLA (Standard priority can take up to ~15 hours). 39. Geographic routing. 40. RPO (Recovery Point Objective) — it's about acceptable data loss, not downtime.

MEMORIZE THIS

If you can answer all forty cold, explain every near-twin in Appendix B, and perform each task once in a real subscription, you're ready to book AZ-104.